Multigranularity Manifold Hybrid Adaptive Near-Neighbor Preserved Clustering for Hyperspectral Image Band Selection

Qingyan Li, Xiaobo Luo, and Yaxu Wang

***Abstract*— Band selection is an effective method to reduce the dimensionality of hyperspectral remote sensing images (HSIs). However, most existing band selection methods usually consider each band as an overall feature and neglect to analyze the intrinsic discriminative components of each band. Worse still, these methods also assume that the spatial features of bands in HSIs are distributed only in Euclidean or non-Euclidean space, which may lead to the limited discriminability of the selected bands. To address these problems, a novel multigranularity manifold hybrid adaptive near-neighbor preserved (MMANP) clustering method is proposed for band selection. Specifically, we first transform the problem of analyzing the intrinsic discriminative components of bands into a feature decoupling problem and propose a fine-grained manifold collaborative (FGMC) feature decoupling model, which can obtain spatially discriminative features and important features with the fine-grained manifold structures in each band. Then, the hybrid adaptive local neighborhood (HALN) block clustering model is proposed to interact with the spatial effective information of the obtained fine-grained important features of each band in the Euclidean and non-Euclidean spaces, which helps to improve the HSI's adaptive clustering of similar bands. Next, multiview cross-granularity learning (MVCL) feature selection model is proposed to harmoniously integrate information at different levels of granularity in HSIs into a unified framework to perform adaptive band selection for each cluster from the perspectives of both different subsets and the whole HSI. Extensive experimental results on four datasets demonstrate that MMANP outperforms other state-of-the-art comparative methods.**

***Index Terms*— Band selection, fine-grained manifold collaborative (FGMC) feature decoupling, hyperspectral images, multigranularity manifold hybrid adaptive near-neighbor preserved (MMANP) clustering, multiview cross-granularity learning (MVCL).**

NOMENCLATURE

Notation	Definition
X	3-D HSI cube.
M, N, and B	Height, width, and the number of bands, respectively.
X^p	First principal component of X.
$\Im_j$	jth segmentation region of X^i.
$\tilde{L}^i$	Normalization pseudo-label of X^i.
L^i	Pseudo-label of X^i.
Z^i	Fine-grained constraint matrix of $\tilde{L}^i$.
L^i	Pseudo-label of X^i.
L_{X^i}	Laplacian matrix of X^i.
$\tilde{W}$	Nonlinear nearest-neighbor similarity matrix of X.
W	Hybrid nearest-neighbor similarity matrix of X.
A	Block diagonal square matrix.
I	Identity matrix.

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I. INTRODUCTION

HYPERSPECTRAL remote sensing is an effective method for observing the Earth and has been used in mapping [1], environmental monitoring [2], and agriculture [3]. Though hyperspectral remote sensing brings a richness of information, it presents the challenge of analyzing high-dimensional data [4]. Considering the strong correlation among neighboring bands of hyperspectral remote sensing images (HSIs), dimensionality reduction becomes a promising data processing method.

To date, the process of dimensionality reduction for HSIs gradually forms two main research ideas: feature extraction [5], [6], [7] and feature selection [8], [9], [10], [11], [12], [13]. Feature extraction involves the dimensional reduction of band information by projecting the original band information into a low-dimensional subspace based on some criteria. The common methods for feature extraction include principal component analysis (PCA) [14], linear discriminant analysis (LDA) [15], and locality preserving projections (LPP) [16]. However, these feature extraction methods typically lose the physical significance of the original space in HSIs, which may limit the generalization performance of the dimensionality-reduced features. In contrast, feature selection focuses on selecting the most representative subset of bands from the original bands as the final dimensionality-reduced results according to certain criteria. Notably, with this approach, the original physical meaning embedded in the selected bands can be preserved as much as possible, which has a positive impact on the generalization performance of this subset. In view of

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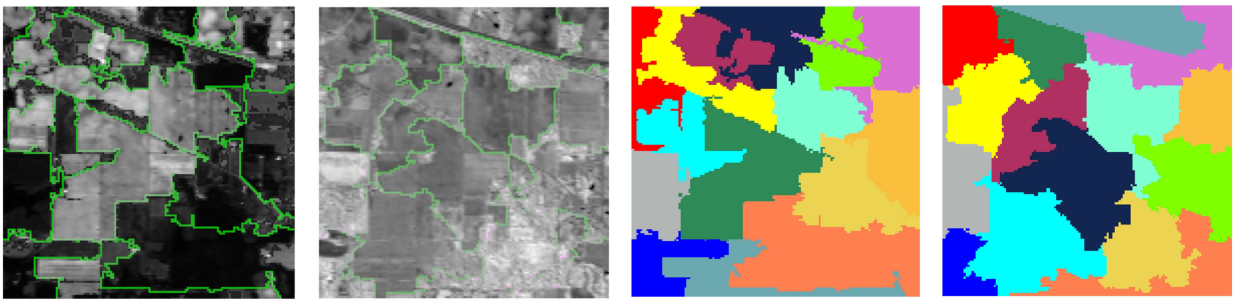


Fig. 1. Differences in spatial structures and intrinsic discriminative features of different bands in the InP image. (a) 30th band in the InP image. (b) 50th band in the InP image. (c) Pseudo-label of the 30th band of the InP image. (d) Pseudo-label of the 50th band of the InP image.

this, this article employs the feature selection method for band selection of HSIs.

According to the availability of class information, the band selection work for HSIs can be classified into supervised learning [17], semi-supervised learning [18], [19], and unsupervised learning [20]. In general, band selection methods (supervised learning and semi-supervised learning) using class information always achieve satisfactory band selection results [21]. However, the cost of obtaining class information on HSIs is high. In other words, the unsupervised band selection methods have more flexible application scenarios. Therefore, this article focuses on unsupervised band selection methods.

In the past decades, unsupervised band selection methods have given rise to three main research approaches: ranking-based methods [20], [22], searching-based methods [23], and clustering-based methods [24], [25], [26]. In the searching-based methods, some scholars have used evolutionary and particle swarm algorithms [27], [28], [29] to select the desired bands. Similarly, the design of the multiobjective unsupervised band selection models also incorporates interband properties. These abovementioned methods can achieve good band selection results. However, these methods have a common problem with computational difficulty.

The ranking-based band selection methods comprise two parts, i.e., band performance evaluation and band ordering. Now, numerous studies have been performed by researchers on band evaluation. Some important criteria such as relative/absolute Boltzmann entropy [30], structural similarity (SSIM) [31], [32], and band sampling [33], [34] are used to explore band selection for HSIs. The clustering-based band selection method also consists of two parts: clustering of band subsets and selection of bands in each subset. Scholars have done much exploration work around band clustering (see [35], [36], [37]). These methods usually perform clustering of bands according to common similarity evaluation metrics (e.g., Euclidean distance, etc.). These ranking-based and clustering-based band selection methods can achieve desirable band selection results and have attracted more attention.

In recent years, a series of excellent results has been achieved in graph-based band selection methods among the abovementioned methods. Zhang et al. [38] extracted the spatial information of homogeneous regions in graph structures through superpixel segmentation and further incorporated

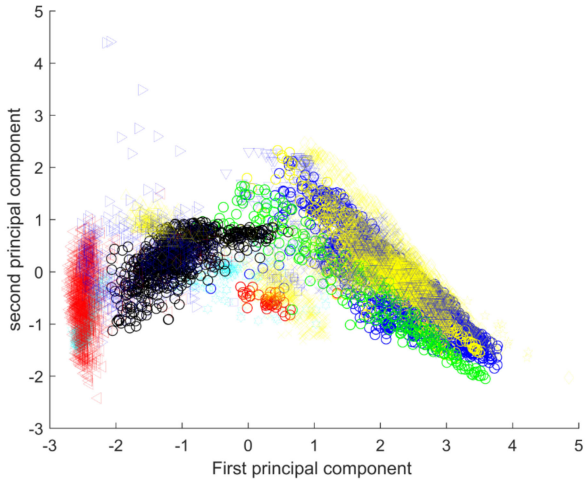


Fig. 2. Scatterplot displays the distribution of samples in the first 2-D space obtained by PCA of the InP image.

the structural information of infinitely corrupted pixels for unsupervised band selection. Wang et al. [39] used a superpixel segmentation algorithm to extract homogeneous regions and constructed similarity graphs for each homogeneous region in HSIs. Then, all potential features are fused to form a unified HSI feature representation. Jia et al. [40] employed superpixel-level distance measurements to measure the similarity among bands and introduced the grouping paradigm to deal with the contextual information among bands. Then, a fast-density clustering method based on the superpixel level was developed for band selection. Although these methods can achieve good band selection results, there are still at least two common problems with these methods. First, these methods always consider each band as an overall feature, and it is difficult to analyze the intrinsic discriminative components of each band. In other words, the differences among bands are reflected both in the spatial structure and the intrinsic discriminative components. Here, some figures in Fig. 1 of spatial structure differences and intrinsic discriminative features of different bands of the Indian Pines (InP) image are given as examples. Therefore, analyzing the intrinsic discriminative components of each band should not be neglected. Second, these methods suffer from the limitation of considering only the spatial features of each band in HSIs to be distributed in

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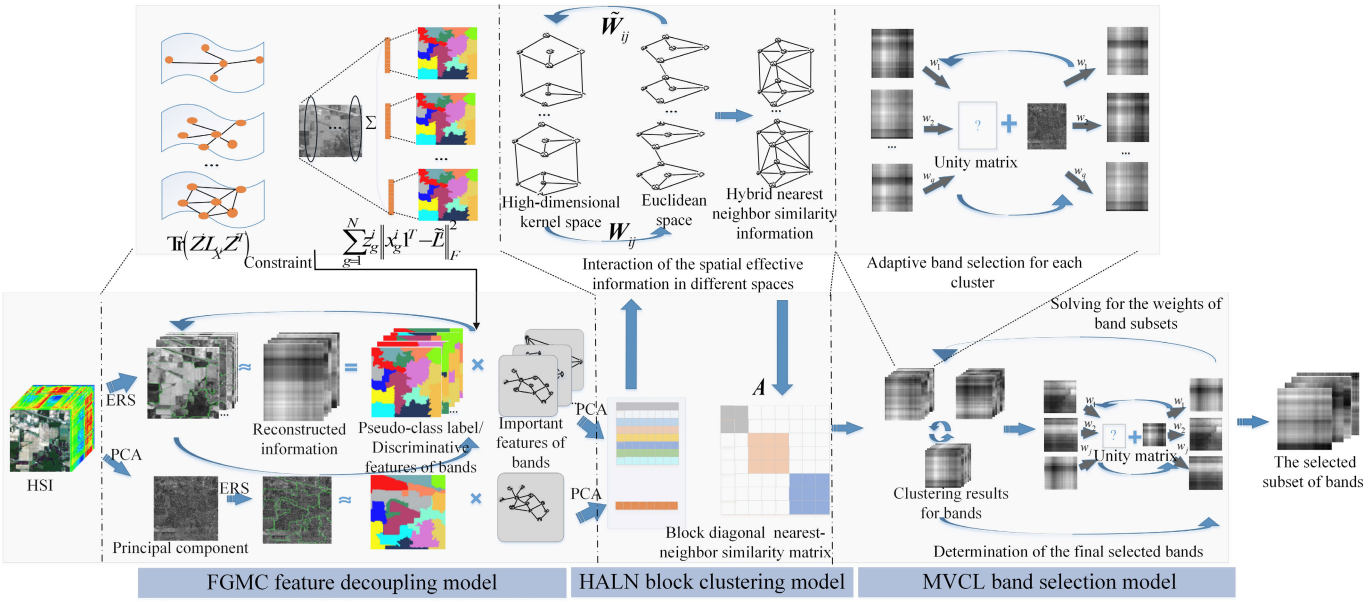


Fig. 3. Framework of the proposed MMANP method.

Euclidean or non-Euclidean space. In fact, the distribution of the ground objects in the real scene is complex, and during the hyperspectral image acquisition process, the distribution of features in HSIs is affected by spectral properties (i.e., high-dimensional properties and information redundancy). Thus, HSIs usually exhibit a nonlinearly discriminative state. It is difficult to describe the real ground object distribution by only using the structural information of Euclidean or non-Euclidean space. Here, the scatterplot in Fig. 2 of the distribution of the samples in the first 2-D space obtained by PCA is given as an example in the InP image. Worse still, this factor may lead to the limited discriminability of the selected bands of HSIs.

In light of this, to address the above two problems, a novel multigranularity manifold hybrid adaptive near-neighbor preserved (MMANP) clustering method is proposed for band selection, as shown in Fig. 3. Unlike previous band selection methods, inspired by the idea of "divide-and-conquer," MMANP first decouples the problem of determining a subset of HSI bands into three subproblems, i.e., feature expressions of bands, clustering of similar bands, and band selection. The above three subproblems are sequentially processed to select band subsets for HSIs. Specifically, we first transform the problem of analyzing the intrinsic discriminative components of bands into a feature decoupling problem and propose a fine-grained manifold collaborative (FGMC) feature decoupling model. This model can obtain spatially discriminative and important features with fine-grained manifold structures for each band. Then, to overcome the limitation of considering only the distribution of spatial features in each band in the Euclidean or non-Euclidean space, the hybrid adaptive local neighborhood (HALN) block clustering model is proposed. This model can take the nonlinear and the linear nearest-neighbor similarity graphs for the fine-grained important features of each band to be considered simultaneously in the unified model to auxiliary the more accurate clustering among similar bands. Next, to overcome the limitation of single-view band selection, a multiview cross-granularity learning (MVCL) feature selection model is proposed to integrate the exploration of the level of importance of different subsets from the perspectives of both the different subsets and the whole HSI. This model can perform adaptive band selection for each cluster. The main contributions of this work are summarized as follows.

1) To the best of authors' knowledge, the FGMC feature decoupling model is the first time that integration of the local geometric structure, local coarse-grained features, and intrinsic fine-grained discriminative features into a unified feature decoupling framework. In this way, the expressions of discriminative and important features with the fine-grained manifold structures of bands can be achieved.
2) The HALN block clustering model is proposed to enhance the adaptive clustering of similar bands in HSIs, ultimately improving the limited discriminability of the final selected bands of HSIs. This model can interact with the spatial effective information of the fine-grained important features of each band in the Euclidean and non-Euclidean spaces simultaneously.
3) The MVCL band selection model is proposed to perform adaptive band selection for each cluster from the perspectives of both different subsets and the whole HSI. This model can harmoniously integrate information at different levels of granularity in HSIs into a unified framework.
4) We design an alternating iteration strategy to solve the objective function, and the experimental results from detailed experiments on four public datasets demonstrate the superiority of the proposed method over some state-of-the-art methods.

The remainder of this article is organized as follows. In Section II, we present a priori work on graph structures. Section III introduces the proposed MMANP clustering model.

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Section IV presents the optimization scheme of the model. Section V reports extensive experiments including comparisons with some state-of-the-art methods. Section VI concludes this article.

II. Prior Works

In this section, to improve the article's readability, some notational conventions are first reported in the Nomenclature. Then, we present algorithms for self-representation learning and the construction of the similarity matrix.

A. Construction of the Similarity Matrix

Constructing similarity graphs as a core problem has received increasing attention in graph-based clustering models [24]. Now, there are two types of similarity graph structures: asymmetric directed graphs and symmetric undirected graphs. Among them, undirected graphs are widely used, thanks to their versatility and flexibility. We are concerned with undirected graphs with local nearest-neighbor learning in this article. The local nearest-neighbor similarity graph is mainly obtained by the following methods: the ε-neighborhood graph, the k-nearest neighbor graph, and the adaptive graph with the Gaussian method. Among them, k-nearest neighbor has been widely used for its ease of implementation. Here, we provide a detailed introduction to the construction of the k-nearest neighbor graph.

Suppose a graph $G = (V, E)$, where samples are regarded as vertexes $v \in V$, relationships among samples are viewed as weighted edges $e_{ij} \in E$. Given any two samples x_i, x_j, the k-nearest neighbor graph can be constructed

$$G_k = \begin{cases} \exp\left(\dfrac{\|x_i - x_j\|^2}{2\sigma^2}\right), & x_i \in \mathcal{K}(x_j), \text{ or } x_j \in \mathcal{K}(x_i) \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $\mathcal{K}(x_j)$ is a set of k-nearest neighbors of x_j. However, the constructions of similarity graphs are highly dependent on the similarity of the data, which may cause the resulting similarity matrix to be sensitive and not flexible enough. To address this problem, Nie et al. [41] proposed a clustering with adaptive neighbor (CAN) model to assign the neighbors for all samples

$$\begin{aligned} &\min \sum_{i,j=1}^{n} \left(\|x_i - x_j\|_2^2 s_{ij} + \gamma s_{ij}^2\right) \\ &\text{s.t. } s_i^T 1 = 1, \quad s_i \geq 0, \ \operatorname{rank}(L_S) = n - c \end{aligned} \tag{2}$$

where $\gamma > 0$ is a regularization parameter. In (2), for any data point x_i, x_j has a probability s_{ij} of being the nearest neighbor to it. With this approach, the resulting s_i is a nearest-neighbor arrangement that preserves the original intrinsic structural information. In recent research work, it has been widely used in various fields, such as face recognition [42], [43] and band selection [11], [44].

B. Self-Representation Learning

Self-representation learning belongs to the unsupervised learning model and has been widely used in graph-based band selection methods. The purpose of self-representation learning is to learn the minimum reconstructed residuals of feature self-expressions. Suppose that X is the data matrix and W is the coefficient matrix. Then, the self-representation model can be expressed as

$$\min_W \mathcal{L}(X - XW) + \theta \mathcal{R}(W) \tag{3}$$

where $\mathcal{L}(X - XW)$ is the loss function, $\mathcal{R}(W)$ is the regularization term, and θ is the balanced parameter. Different learning models in (3) can be achieved by adjusting the loss function and the regularization term [11], [38].

In recent years, self-representation models have been widely used in HSI band selection. Zhang et al. [45] incorporated spatial similarity graphs and band similarity graphs from the homogeneous regions obtained via superpixel segmentation into the self-representation model for band selection. Similarly, Tang et al. [46] incorporated band correlation, spatial-region similarity graphs, and weighted principal component information into a self-representation model to select the most informative subset of bands for HSIs. Furthermore, to mitigate the effect of noise in the bands on the similarity graph, Wang et al. [44] introduced the adaptive neighbor arrangement term in Section II-A into the self-representation model for band selection. Although these methods produce good selected bands, they have at least two common problems. First, these methods have problems in analyzing the intrinsic discriminative components of each band as they always consider the band as an overall feature. Second, these methods are limited to exploring the spatial structure information of HSIs in Euclidean or non-Euclidean space.

Unlike these methods, our approach first transforms the problem of analyzing the intrinsic discriminative components of bands into a problem of decoupling discriminative and important features of bands. Besides, our method considers the local spatial adaptive neighborhood structures for the obtained important features of each band in both non-Euclidean and Euclidean spaces simultaneously. Instead of using the structure graph of each band in the Euclidean or non-Euclidean space as a regularization term, our method unifies the obtained local neighborhood structure information of each band in the non-Euclidean and Euclidean spaces in a unified framework to solve the hybrid local adaptive neighborhood graph for each band. We improve the clustering of similar bands in HSIs by interacting with the spatial effective information of the fine-grained important features of each band in the Euclidean and non-Euclidean spaces.

III. Methodology

This section presents the proposed MMANP algorithm to perform band selection for HSIs. MMANP consists of three parts: the FGMC feature decoupling model, the HALN block clustering model, and the MVCL band selection model. The flowchart of the MMANP algorithm is shown in Fig. 4. The

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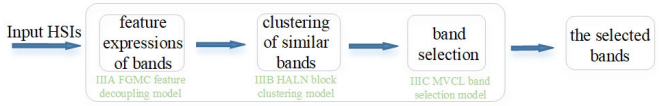


Fig. 4. Flowchart of the MMANP algorithm.

detailed theory of each part is presented in this section. Overall, MMANP first decouples bands to obtain the fine-grained discriminative and important features for each band while considering the spatial manifold information. Then, the HALN matrices of the resulting fine-grained important features and their block diagonal similarity matrices are solved for the clustering of similar bands. Next, MVCL is employed to perform the most discriminative band selection on the obtained subsets of bands.

A. FGMC Feature Decoupling Model

Given the HSI cube $X \in \Re^{M\times N\times B}$, B is the number of bands and $M \times N$ is the size of the scene. From the information processing perspective, both $X = \{X^1, X^2, \ldots, X^B\}$ and X^p are coarse-grained features of X, where X^p is the first principal component of X and X^i is the ith band of X. To improve the accuracy of feature representation for X, $\{X^1, X^2, \ldots, X^B\}$ and X^p are performed further fine-grained feature descriptions, respectively. The details are as follows.

1) Delineations of Spatially Homogeneous Regions: Given a certain ground scene, different classes in that scene typically have different spectral properties, while the same classes have similar spectral properties. In light of this, we utilize the entropy rate superpixel segmentation (ERS) method [47] to partition regions of spatial homogeneity for $\{X^1, X^2, \ldots, X^B\}$ and X^p, respectively. Then, we can obtain S regions by using ERS on $\{X^1, X^2, \ldots, X^B\}$ and X^p, which can be denoted as follows:

$$X^i = \bigcup_j^S \Im_j, \quad \text{s.t. } \Im_j \cap \Im_q = \emptyset. \ (j \neq q). \tag{4}$$

Here, $\Im_j$ is the jth segmentation region. Next, the obtained segmentation results of $\{X^1, X^2, \ldots, X^B\}$ and X^p are the fine-grained features of X.

2) FGMC Feature Decoupling: Unlike the conventional idea of feature decoupling, our FGMC feature decoupling model focuses on separating the discriminative and important features of X^i, $i \in \{1, 2, \ldots, B, p\}$. Specifically, we first define the normalization of the pseudo-label $\tilde{L}^i$ corresponding to the fine-grained feature X^i as the fine-grained discriminative feature of X^i, i.e., $\tilde{L}^i = L^i/\max(L^i)$. Here, L^i is the pseudo-label of X^i and $\max(L^i)$ is the number of pseudo-labels. To reflect the intrinsic relationships of fine-grained discriminative features as accurately as possible, then, we design a new feature decoupling model depending on (4), which is defined as

$$\min_{Z^i} \left\| X^i - \tilde{L}^i Z^i \right\|_{2,1}. \tag{5}$$

Here, Z^i is the fine-grained constraint matrix of $\tilde{L}^i$, which is called the fine-grained important features of X^i. Next, to project a "good" map of the original high-dimensional space in X^i in the fine-grained low-dimensional important feature space, a reasonable graph regularization term is derived. By incorporating the graph regularization term into (5), it can be rewritten as

$$\min_{Z^i} \left\| X^i - \tilde{L}^i Z^i \right\|_{2,1} + \mathrm{Tr}\left(Z^i L_{X^i} Z^{i\mathrm{T}}\right). \tag{6}$$

In (6), $L_{X^i} = D_{X^i} - W_{X^i}$ is the Laplacian matrix of X^i, where W_{X^i} is the similarity matrix of X^i, which can be solved by the adaptive and optimal neighbors for each pixel according to the local distances [41]. With this approach, the potential manifold structure implied in the original X can be preserved in Z^i as much as possible. Furthermore, to accurately describe the local fine-grained important features of bands, a novel idea of cooperative representation (CF) is proposed.

Definition 1: Without loss of generality, the intrinsic fine-grained discriminative feature L_{X^i} of X^i is composed of all local coarse-grained features of X^i.

According to this definition, the proposed novel CF model can be expressed as follows:

$$\tilde{\mathcal{R}} = \sum_{g=1}^{N} z_g^i \left\| \left(\tilde{L}^i\right)^{-1} \left(x_g^i 1^{\mathrm{T}}\right) \right\|_F^2. \tag{7}$$

The computational complexity of (7) is high because it involves multiple multiplication calculations. In view of this, (7) can be relaxed as

$$\mathcal{R} = \sum_{g=1}^{N} z_g^i \left\| x_g^i 1^{\mathrm{T}} - \tilde{L}^i \right\|_F^2. \tag{8}$$

The above constraint adaptively incurs the penalty z_g^i according to the differences between x_g^i and $\tilde{L}^i$. Thus, by minimizing (8), our intuition is that x_g^i should be as close as possible to the fine-grained discriminative feature $\tilde{L}^i$, whose corresponding penalty z_g^i also tends to be as small as possible.

Moreover, by incorporating the proposed equation (8) into (6), the FGMC feature decoupling model for X^i is proposed, which can be rewritten as

$$\min_{Z^i} \left\| X^i - \tilde{L}^i Z^i \right\|_{2,1} + \beta \mathrm{Tr}\left(Z^i L_{X^i} Z^{iT}\right) + \lambda \left(\sum_{g=1}^{N} z_g^i \left\| x_g^i 1^{\mathrm{T}} - \tilde{L}^i \right\|_F^2 \right). \tag{9}$$

The β and $\lambda \geq 0$ are the regularization parameters. Here, the size of $\tilde{L}^i$ is $M \times N$. With the size of $N \times N$, the fine-grained important feature $Z^i \geq 0$ is employed to capture the local geometric structure of X^i while further describing the potential linear constraints between the local coarse-grained features x_g^i and the intrinsic fine-grained discriminative features $\tilde{L}^i$ of X^i. Instead of simply using the feature decomposition, the FGMC feature decoupling model of (9) explicitly presents the concepts of discriminative and important features, and further jointly integrates local geometric structure, local coarse-grained features, and intrinsic fine-grained discriminative feature information in the unified feature decoupling framework. In this way, it facilitates the

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effective representation of discriminative and important features with the fine-grained manifold structures for each band.

B. HALN Block Clustering Model

The detailed information on the spatial features of bands can be reflected in the fine-grained important features in each band of HSIs. To analyze the similarity among the obtained fine-grained important features of different bands in Section III-A, we propose the HALN block clustering model. Unlike local adaptive neighborhood models based on Euclidean space [41], our proposed HALN block clustering model focuses on constructing hybrid nearest-neighbor similarity graphs for each band by simultaneously considering the distribution characteristics of spatial features of HSIs in both Euclidean and non-Euclidean spaces. The details are as follows.

1) Solving the Nonlinear Nearest-Neighbor Similarity Matrix: The most popular method for extracting nonlinear features from a scene is to use the kernel method [48]. Given the fine-grained important feature Z^i of X^i, $i \in \{1, 2, \ldots, B, p\}$, the Z^i is first projected to high-dimensional kernel space $\aleph$, i.e., $\phi(Z^i)$. Here, $\phi : Z^i \in \Re \to \phi(Z^i) \in \aleph$, and kernel matrix corresponding to X is $Z_K \in \aleph^{N\times N\times B}$; it can be calculated by

$$Z_K = \begin{bmatrix} k\big(\big(Z^1 .* Z^{1\mathrm{T}}\big) * \big(Z^1 .* Z^{1\mathrm{T}}\big), Z_1^p\big) \\ k\big(\big(Z^2 .* Z^{2\mathrm{T}}\big) * \big(Z^2 .* Z^{2\mathrm{T}}\big), Z_1^p\big), \ldots \\ k\big(\big(Z^B .* Z^{B\mathrm{T}}\big) * \big(Z^B .* Z^{B\mathrm{T}}\big), Z_1^p\big) \end{bmatrix}. \tag{10}$$

Here, the kernel function uses the polyplus kernel function, i.e.,

$$\begin{aligned} &k\big(\big(Z^1 .* Z^{1\mathrm{T}}\big) * \big(Z^1 .* Z^{1\mathrm{T}}\big), Z_1^p\big) \\ &\quad = \big(\big(Z^1 .* Z^{1\mathrm{T}}\big) * \big(Z^1 .* Z^{1\mathrm{T}}\big) \times Z_1^p + 1\big)^2. \end{aligned} \tag{11}$$

Suppose that the local neighborhood space of each band in the kernel space is a homogeneous Euclidean space. Then, to learn the band similarity matrix W_K^i for Z_K, the objective function can be expressed as

$$\min_{\substack{0\le W_K^i\le 1 \\ 1^{\mathrm{T}}W_K^i=1}} \sum_{i,j=1}^{B} \left\| Z_K^i - Z_K^j \right\|_2^2 W_K^{ij} + \delta \sum_{i}^{B} \left\| W_K^i \right\|_2^2. \tag{12}$$

To capture the potential nonlinear features while performing dimensionality reduction of Z_K^i in (12), PCA is employed in the high-dimensional fine-grained constraint space to capture potential nonlinear relationships in the original fine-grained constraint features as much as possible, i.e.,

$$\tilde{z}_i = P^1 Z_K^i. \tag{13}$$

Thus, (12) can be rewritten as

$$\min_{\substack{0\le \tilde{w}_i\le 1 \\ 1^{\mathrm{T}}\tilde{w}_i=1}} \sum_{i,j=1}^{B} \left\| \tilde{z}_i - \tilde{z}_j \right\|_2^2 \tilde{w}_{ij} + \delta \sum_{i}^{B} \left\| \tilde{w}_i \right\|_2^2. \tag{14}$$

The $\delta \ge 0$ is a regularization parameter. By solving (14), the nonlinear nearest-neighbor similarity matrix $\tilde{W}$ among the spatial features of bands in HSIs can be obtained. However, the distribution of the ground objects in the real scene is complex, which may include both nonlinear and linear distributions of spatial features. Therefore, both nonlinear and linear nearest-neighbor relationships among the spatial features of bands need to be considered when determining the nearest-neighbor similarity graphs among bands.

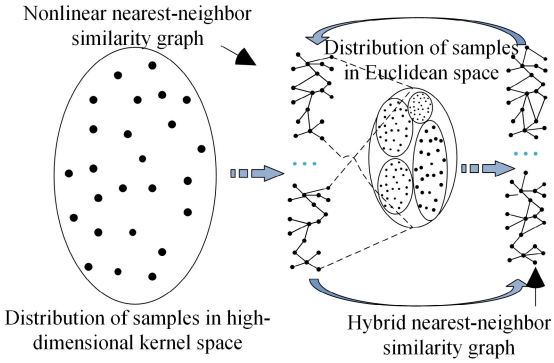


Fig. 5. Flowchart for solving the hybrid nearest-neighbor similarity matrix.

2) Solving the Hybrid Nearest-Neighbor Similarity Matrix: Furthermore, to simultaneously solve the nonlinear and linear nearest-neighbor similarity graphs among the spatial features of bands, a new solution algorithm is proposed, and the flowchart is shown in Fig. 5.

The expression for the hybrid nearest-neighbor similarity matrix is defined as

$$\begin{aligned} \min_{\tilde{w}_i, w_i} &\sum_{i,j=1}^{B} \left\| \tilde{z}_i - \tilde{z}_j \right\|_2^2 \tilde{w}_{ij} + \delta \sum_{i}^{B} \left\| \tilde{w}_i \right\|_2^2 \\ &+ \sum_{i,j=1}^{B} \left\| \tilde{w}_i \hat{z}_i - \tilde{w}_i \hat{z}_j \right\|_2^2 w_{ij} + \varepsilon \sum_{i=1}^{B} w_{ij}^2 \\ \text{s.t. } & 1^{\mathrm{T}}\tilde{w}_i = 1, \quad 1^{\mathrm{T}}w_i = 1, \ 0 \le \tilde{w}_{ij} \le 1 \\ & 0 \le w_{ij} \le 1, \quad \tilde{w}_{ii} = 0. \end{aligned} \tag{15}$$

Similarly, $\hat{z}_i$ is the fine-grained important feature obtained by PCA in (15), i.e., $\hat{z}_i = P^1(Z^i * Z^{iT})$. Here, the different i are independent of each other. Thus, (15) can be rewritten as

$$\begin{aligned} \min_{\tilde{w}_i, w_i} &\sum_{i=1}^{B} \left\| \tilde{z}_i - \tilde{z}_j \right\|_2^2 \tilde{w}_{ij} + \delta \left\| \tilde{w}_i \right\|_2^2 \\ &+ \sum_{i=1}^{B} \left\| \tilde{w}_i \hat{z}_i - \tilde{w}_i \hat{z}_j \right\|_2^2 w_{ij} + \varepsilon \sum_{i=1}^{B} w_{ij}^2 \\ \text{s.t. } & 1^{\mathrm{T}}\tilde{w}_i = 1, \quad 1^{\mathrm{T}}w_i = 1, \ 0 \le \tilde{w}_{ij} \le 1 \\ & 0 \le w_{ij} \le 1, \quad \tilde{w}_{ii} = 0. \end{aligned} \tag{16}$$

The $\varepsilon \ge 0$ is a regularization parameter. As shown in (16), the nonlinear and linear nearest-neighbor similarity graphs among the spatial features of bands are harmoniously integrated into the unified model. This idea is beneficial to capture the intrinsic relationship between the distributions of the spatial features reflected by different bands of HSIs in real feature scenes as efficiently as possible. With this approach, the spatial effective information of band distributions in Euclidean and non-Euclidean spaces is used interactively, which helps to improve the discriminative ability to capture

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similar bands in HSIs. Considering the strong correlation among the nearest neighbor bands of HSIs, we further solve the block diagonal square matrix A corresponding to the obtained hybrid nearest-neighbor similarity matrix W.

3) Solving Block Diagonal Matrix for the Hybrid Nearest-Neighbor Similarity Matrix: Next, inspired by Li et al. [37], A can be formulated as

$$\min_{\sum_j a_{ij}=1, a_{ij}\geq 0} \sum \|A - W\|_2^2 + 2\mu Tr\left(F^{\mathrm{T}} L_a F\right). \tag{17}$$

Furthermore, (17) can be rewritten as

$$\min_{\sum_j a_{ij}=1, a_{ij}\geq 0} \sum_j \left(a_{ij} - w_{ij}\right)^2 + \mu \sum_j \left\|f_i - f_j\right\|_2^2 a_{ij}. \tag{18}$$

By denoting $o_{ij} = \|f_i - f_j\|_2^2$ and o_i is a vector with the jth element in o_{ij}, (18) can be simplified as

$$\min_{a_i^T 1=1, a_i\geq 0} \left\|a_i - \left(w_i - \frac{\mu}{2} o_i\right)\right\|_2^2. \tag{19}$$

Here, $\mu = 0.1$ [37]. According to karush-kuhn-tucker (KKT) condition, η_i can be calculated as $\eta_i = (2/m) - ((2\sum_{j=1}^m (w_i - (\mu/2)o_i))/m)$. Finally, a_{ij} can be denoted as

$$a_i = \left(w_i - \frac{\mu}{2} o_i + \frac{1}{m} - \frac{\sum_{j=1}^m \left(w_i - \frac{\mu}{2} o_i\right)}{m}\right)_+. \tag{20}$$

Unlike the method of Li et al. [37], the block diagonal nearest-neighbor similarity matrix A can be adaptively extended by using the identity matrix so that the dimension of A becomes $\max(M, N) \times \max(M, N)$. In this way, the computation can be easier. Besides, the clustering result of the band subset is determined according to the block clustering result of A. Furthermore, the final selected bands are determined by the obtained clustering results for each subset of bands.

C. MVCL Band Selection Model

To date, most band selection methods prefer to apply some techniques (e.g., information entropy) to the clustering results to determine the selected bands. However, these methods always give a selected subset of bands from a single viewpoint, which can make the selected bands limited. To alleviate this problem, numerous band selection methods based on the idea of multiview have been proposed. However, most of them have difficulty in effectively utilizing different granularity features simultaneously. In view of this, the MVCL band selection model is proposed. Specifically, we first perform feature enhancement for each band using the fine-grained important and discriminative features of each band, i.e.,

$$\tilde{X}^i \approx \tilde{L}^i Z^i + X^i \tag{21}$$

where $\tilde{X}^i$ is the enhanced feature of X^i. Then, we design the MVCL model to measure the importance of different band subsets of $\tilde{X}^i$ for HSIs.

1) Solving for the Weights of Band Subsets: Mathematically, this problem can be modeled as follows:

$$\begin{aligned} &\min_{w_v, \tilde{P}_f} \sum_{v=1}^{c} w_v \left(\left\|\tilde{P}_f - P_1^v\right\|_2^2 + \tau \left\|\tilde{P}_f - P_A\right\|_2^2\right) \\ &\text{s.t. } \tilde{P}_f \geq 0, \quad 1^{\mathrm{T}} \tilde{P}_f = 1 \end{aligned} \tag{22}$$

where c is the number of band clusters, P_1^v is the first principal component corresponding to the vth subset, P_A is the first principal component corresponding to all $P_1^v, v = \{1, 2, \ldots, c\}$, w_v is the weight matrix corresponding to all the subsets, and $\tilde{P}_f$ is the unity matrix corresponding to the first principal component of all the subsets. Different levels of granularity information in bands of HSIs are harmoniously integrated into a unified framework in (22). It is beneficial to integrate the exploration of the level of importance of different subsets from the perspectives of both different subsets and the whole HSI. Similarly, the weights corresponding to different bands in each subset can be solved by (22).

2) Determination of the Final Selected Bands: To further determine the numbers of selected bands for subsets of the HSI, the weight percentages of subsets are calculated separately, i.e.,

$$p_c = w_{P_1^v} \Big/ \sum_{v=1}^{c} w_{P_1^v}. \tag{23}$$

Finally, the bands of each subset are sorted in descending order according to the obtained weights, and the bands of the top $p_c o$ are selected as the selected bands of each subset, respectively. Here, o is the number of final selected bands for HSIs.

IV. Optimization Algorithms

Furthermore, this section presents the optimization process for Sections III-A–III-C models in Sections IV-A–IV-C, respectively. The details are as follows.

A. Optimization Algorithm for (9)

According to some algebraic steps, we can rewrite (9) as the following objective function:

$$\begin{aligned} \min_{Z^i} &\left\|X^i - \tilde{L}^i Z^i\right\|_{2,1} + \beta \mathrm{Tr}\left(Z^i L_{X^i} Z^{iT}\right) \\ &+ \lambda \sum_{g=1}^{N} \left\|\left(x_g^i 1^{\mathrm{T}} - \tilde{L}^i\right)\kappa_{1/2}^i\right\|_F^2 \end{aligned} \tag{24}$$

where $\kappa^i = \mathrm{diag}(z^i)$. To solve (24), we relax $\|X^i - \tilde{L}^i Z^i\|_{2,1}$ by $\mathrm{Tr}((X^i - \tilde{L}^i Z^i)^{\mathrm{T}} C_X (X^i - \tilde{L}^i Z^i))$; here, C_X is a diagonal matrix, i.e.,

$$C_{X_{ii}} = \left(1 \Big/ \left(2\left\|x^i - \tilde{L}^i z^i\right\|_2 + \tilde{\varepsilon}\right)\right). \tag{25}$$

Here, $\tilde{\varepsilon}$ is a constant as small as possible. Therefore, (24) can be reformulated as

$$\begin{aligned} J\left(Z^i\right) = &\mathrm{Tr}\left(\left(X^i - \tilde{L}^i Z^i\right)^{\mathrm{T}} C_X \left(X^i - \tilde{L}^i Z^i\right)\right) \\ &+ \lambda \mathrm{Tr}\left(\sum_{i=1}^{N} \left(\left(x_g^i 1^{\mathrm{T}} - \tilde{L}^i\right)\kappa^i \left(x_g^i 1^{\mathrm{T}} - \tilde{L}^i\right)^{\mathrm{T}}\right)\right) \\ &+ \beta R. \end{aligned} \tag{26}$$

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Algorithm 1 Optimization for FGMC Feature Decoupling Model

Input: Hyperspectral dataset X^i, $i \in \{1, 2, \ldots, B, p\}$, the pseudo-label L^i of X^i, parameters β, λ;
Output: The fine-grained constraint matrix Z^i of $\tilde{L}^i$;
1. Construct the normalization of the pseudo-label $\tilde{L}^i$ of X^i ;
2. Calculate the Laplacian matrix L_{X^i} of X^i ;
3. Construct the auxiliary matrices C_1 and D_1;
repeat
4. Update the fine-grained constraint matrix Z^i by Eq. (29);
5. Update the matrix C_X by Eq. (25);
until the maximum iteration reached.
6. **return** The fine-grained constraint matrix Z^i of $\tilde{L}^i$.

Here, $R = \beta \mathrm{Tr}(Z^i L_{X^i} Z^{iT})$. To solve (26), the derivative with respect to Z^i is calculated by

$$\frac{\partial J}{\partial Z^i} = -2X^{iT} C_X \tilde{L}^i + 2\tilde{L}^{iT} Z^i C_X \tilde{L} + \lambda \sum_{i=1}^{N} \left(C_1 - 2\tilde{L}X^i + D_1\right) + 2\beta L_{X^i} Z^i = 0. \quad (27)$$

According to (27), the expression for Z^i can be given by

$$Z^i = \tilde{Q}^{-1}\left(2X^{iT} C_X \tilde{L} - \lambda\left(C_1 - 2\tilde{L}X^i + D_1\right)\right). \quad (28)$$

Here, $\tilde{Q} = 2\tilde{L}^{iT} C_X \tilde{L}^i + 2\beta L_{X^i}$ and Q equals $\tilde{Q} + I$ to avoid the problem of $\tilde{Q}$ being a singular matrix, i.e.,

$$Z^i = Q^{-1}\left(2X^{iT} C_X \tilde{L} - \lambda\left(C_1 - 2\tilde{L}X^i + D_1\right)\right). \quad (29)$$

Besides, here also define column vector $c_1 = \mathrm{diag}(X^{iT} X^i)$ and $C_1 = (c_1, \ldots, c_1)^{\mathrm{T}}$, and the column vector $d_1 = \mathrm{diag}(\tilde{L}^{iT} \tilde{L}^i)$ and $D_1 = (d_1, \ldots, d_1)^{\mathrm{T}}$ [49]. In summary, (29) can be easily solved for the optimal Z^i. Here, the matrix C_X can be initialized with the identity matrix, and the calculation of C_X relies on $\|x^i - \tilde{L}^i z^i\|_{2,1}$. In (29), the C_X and Z^i are alternately updated in each iteration. What is more, the obtained optimal Z^i is the optimal solution of the FGMC feature decoupling model. The algorithmic procedure for the FGMC feature decoupling model is shown in Algorithm 1.

B. Optimization Algorithm for (16)

In (16), this problem can be decomposed into two subproblems to be solved, respectively.

1) Subproblem 1: Fix W_{ij} and Update $\tilde{W}_{ij}$: According to (16), the objective function for subproblem 1 can be simplified to

$$\min_{\tilde{w}_{ij}} \sum_{j=1}^{B} \left\|\tilde{z}_i - \tilde{z}_j\right\|_2^2 \tilde{w}_{ij} + \delta \left\|\tilde{w}_i\right\|_2^2 + \sum_{j=1}^{B} \left(\left\|\tilde{w}_i \hat{z}_i - \tilde{w}_i \hat{z}_j\right\|_2^2 w_{ij}\right) \quad \text{s.t. } 1^{\mathrm{T}} \tilde{w}_i = 1, \quad 0 \le \tilde{w}_{ij}, \ \tilde{w}_{ii} = 0. \quad (30)$$

To improve the readability of this article, we introduce the detailed solution steps of $\tilde{w}_{ij}$ in the Appendix. Here, let $\hat{Z}_{ij} = (\hat{z}_i - \hat{z}_j)^{\mathrm{T}}(\hat{z}_i - \hat{z}_j)$, and the optimal solution of $\tilde{w}_{ij}$ can be denoted as

$$\tilde{w}_{ij} = \begin{cases} \dfrac{d_{i,k+1} - d_{ij}}{-\sum_{h=1}^{k} d_{ih} + k d_{i,k+1} - 2\sum_{h=1}^{k} w_{ih} \hat{Z}_{ij} + 2w_{ij} \hat{Z}_{ij}}, & j \le k \\ 0, & j > k. \end{cases} \quad (31)$$

2) Subproblem 2: Fix $\tilde{W}_{ij}$ and Update W_{ij}: With the above matrix $\tilde{w}_{ij}$, here, the objective function for subproblem 2 of (16) can be defined as

$$\min_{w_{ij}} \sum_{j=1}^{B} \left\|\tilde{W} z_i - \tilde{W} z_j\right\|_2^2 w_{ij} + \varepsilon \sum_{j=1}^{B} w_{ij}^2 \quad \text{s.t. } 1^{\mathrm{T}} \tilde{w}_i = 1, \quad 0 \le \tilde{w}_i. \quad (32)$$

Similarly, let $\hat{d}_{ij} = \|\tilde{W} z_i - \tilde{W} z_j\|_2^2$, and (32) can be rewritten as

$$\min \frac{1}{2}\left\|w_i + \hat{d}_i / 2\varepsilon\right\|_2^2 \quad \text{s.t. } 1^{\mathrm{T}} w_i = 1, \quad 0 \le w_{ij} \le 1. \quad (33)$$

Similar to the solution of (30), the optimal solution of w_{ij} can be formulated as

$$w_{ij} = \begin{cases} \dfrac{\hat{d}_{i,k+1} - \hat{d}_{ij}}{k\hat{d}_{i,k+1} - \sum_{h=1}^{k} \hat{d}_{ih}}, & j \le k \\ 0, & j > k. \end{cases} \quad (34)$$

Besides, the algorithmic procedure for the HALN block clustering model is shown in Algorithm 2.

Algorithm 2 Optimization for HALN Block Clustering Model

Input: The fine-grained constraint matrix Z^i of $\tilde{L}^i$, the first principal component P^1 of Z^i, parameters $\delta, \varepsilon = 1$;
Output: The nonlinear nearest-neighbor similarity matrix $\tilde{W}$ of X, the hybrid nearest-neighbor similarity matrix W of X, block diagonal square matrix A;
1. Construct kernel matrix Z_K^i by Eq. (10) and use Eq. (13) to reduce the dimensionality of Z_K^i;
2. **step1**: Solving of W by Eq. (16);
repeat
3. Fix W, update the $\tilde{W}$ by Eq. (31);
4. Fix $\tilde{W}$, update the W by Eq. (34);
until the maximum iteration reached.
5. **step2**: Solving of A by Eq. (20);
6. **return** $\tilde{W}$, W, A.

C. Optimization Algorithm for (22)

In (22), it is not easy to solve for W_v and $\tilde{P}_f$ simultaneously. With this in mind, let W_v be fixed [50]. Then, (22) can be simplified as

$$\min_{\tilde{P}_f} \sum_{v=1}^{c} \left\|\tilde{P}_f - P_1^v\right\|_2^2 + \tau \left\|\tilde{P}_f - P_A\right\|_2^2 \quad \text{s.t. } \tilde{P}_f \ge 0, \quad 1^{\mathrm{T}} \tilde{P}_f = 1. \quad (35)$$

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The Lagrangian function of (35) is

$$\mathcal{L}(\tilde{p}_f^i, \phi, \varphi) = \frac{1}{2}\sum_{v=1}^{c}\left\|\tilde{p}_f^i - p_1^{vi}\right\|_2^2 + \frac{\tau}{2}\left\|\tilde{p}_f^i - p_A^i\right\|_2^2 - \phi\left(1^{\mathrm{T}}\tilde{p}_f{}^i - 1\right) - \varphi\tilde{p}_f^i \tag{36}$$

where ϕ and φ are the Lagrange multipliers. According to KKT conditions, i.e.,

$$\begin{cases} \forall j, & \sum_{v=1}^{c}\tilde{p}_f^{ij} - \sum_{v=1}^{c}\tilde{p}_1^{vi} - \tau\tilde{p}_A^{ij} + \tau\tilde{p}_f^{ij} - \phi^* - \varphi_j^* = 0 \\ \forall j, & \tilde{p}_f^{ij} \geq 0 \\ \forall j, & \varphi_j^* \geq 0 \\ \forall j, & \tilde{p}_f^{ij}\varphi_j^* = 0. \end{cases} \tag{37}$$

Since $1^{\mathrm{T}}\tilde{p}_f^{i*} = 1$, some mathematical expressions can be obtained, i.e., $\phi^* = ((c - \sum_{v=1}^{c}1^{\mathrm{T}}\tilde{p}_1^{vi} - \tau 1^{\mathrm{T}}\tilde{p}_A^{ij} + \tau - 1^{\mathrm{T}}\varphi^*)/N)$ and $p_f^{i*} = (\sum_{v=1}^{c}\tilde{p}_1^{vi}/(c+\tau)) + (\tau p_A^i/(c+\tau)) + (c/N(c+\tau))1 - (\sum_{v=1}^{c}1^{\mathrm{T}}\tilde{p}_1^{vi}1/N(c+\tau)) - (1^{\mathrm{T}}p_A^i 1/N(c+\tau)) + (1/N(c+\tau)) - (1^{\mathrm{T}}\varphi^* 1/N(c+\tau)) + (\phi^*/(c+\tau))$. Here, let $y = (\sum_{v=1}^{c}\tilde{p}_1^{vi}/(c+\tau)) + (\tau p_A^i/(c+\tau)) + (c/N(c+\tau))1 - (\sum_{v=1}^{c}1^{\mathrm{T}}\tilde{p}_1^{vi}1/N(c+\tau)) - (1^{\mathrm{T}}p_A^i 1/N(c+\tau)) + (1/N(c+\tau))$, $\tilde{\varphi}^* = (1^{\mathrm{T}}\varphi^*/N(c+\tau))$, and the $p_f^{i*} = y - \tilde{\varphi}^* 1 + (\phi^*/(c+\tau))$ can be obtained. Thus, for p_f^{ij*}, we have

$$p_f^{ij*} = y_j - \tilde{\varphi}^* + \frac{\phi_j^*}{c+\tau}. \tag{38}$$

According to (37) and (38), the $y_j - \tilde{\varphi}^* + (\phi_j^*/(c+\tau)) = (y_j - \tilde{\varphi}^*)_+$, where $(\cdot)_+ = \max(\cdot, 0)$, and the optimal solution of p_f^{ij*} can be represented, i.e., $p_f^{ij*} = (y_j - \tilde{\varphi}^*)_+$. Similarly, $\phi_j^* = c + \tau(\tilde{\varphi}^* - y_j)_+$. According to (37), the optimal solution of $\tilde{\varphi}^*$ can be formulated by $\tilde{\varphi}^* = (1/N)\sum_{j=1}^{n}(\tilde{\varphi}^* - y_j)_+$. Here, we define the function of $\tilde{\varphi}$

$$f(\tilde{\varphi}) = \frac{1}{2N^2}\sum_{j=1}^{N}(\tilde{\varphi} - y_j)_+^2 - N\tilde{\varphi}. \tag{39}$$

Let $f(\tilde{\varphi}) = 0$, $\tilde{\varphi}^*$ can be solved by finding the root. Since $\tilde{\varphi} \geq 0$, $f(\tilde{\varphi})$ is the convex function, and Newton's method is used to solve the root of $f(\tilde{\varphi}) = 0$, i.e.,

$$\tilde{\varphi}_{t+1} = \tilde{\varphi}_t - \frac{f(\tilde{\varphi}_t)}{f'(\tilde{\varphi}_t)}. \tag{40}$$

With this approach, $\tilde{P}_f$ can be calculated and fixed $\tilde{P}_f$ to update w_v. According to the work of Wang et al. [50], w_v can be expressed as

$$w_v = \frac{1}{2\sqrt{\left\|\tilde{P}_f - P_1^v\right\|_2^2 + \tau\left\|\tilde{P}_f - P_A\right\|_2^2}}. \tag{41}$$

The algorithmic procedure for the MVCL band selection model is shown in Algorithm 3.

Algorithm 3 Optimization for MVCL Band Selection Model

Input: The 1st principal component corresponding to the vth subset P_1^v of X^i, P_A is the 1st principal component corresponding to all P_1^v, the number of bands c, parameter $\tau = 0.1$;
Output: The weight matrix corresponding to all the subsets W_v, the unity matrix corresponding to the first principal component of all the subsets $\tilde{P}_f$, the selected bands;
repeat
1. Fix W_v, update the $\tilde{P}_f$ by Eqs. (38), (39), (40);
2. Fix $\tilde{P}_f$, update the W_v by Eq. (41);

until the maximum iteration reached.
3. Solving the weight percentages of the subsets by Eq. (23) to select the band subset;
4. **return** W_v, $\tilde{P}_f$, the selected bands.

TABLE I
TIME COMPLEXITY OF DIFFERENT METHODS

Method	Time Complexity
S4P	$\mathcal{O}(B^2MN + MNlog(MN) + TB^3 + S)$
GRSC	$\mathcal{O}(MNlog(MN) + Bo^2 + (S+1)B^3 + (B^3 + SDk)T)$
HLFC	$\mathcal{O}(MNlog(MN) + 2B^3 + Bo^2 + S(m+1))$
FNGBS	$\mathcal{O}(B^2MN + B^3)$
RDGSR	$\mathcal{O}((MN)^2B + (B^2MN + B^3)T)$
RLFFC	$\mathcal{O}(BD + 2Bd^2 + Bo^2 + (S+1)D^3 + MNlog(MN))$
EGCSR	$\mathcal{O}(B^2MN + B^2logB)$
MMANP	$\mathcal{O}(B^3 + 2N^2s + N^2M + N^2c)T$

D. Computational Complexity Analysis

The computational complexity of MMANP consists of three main components, namely, the FGMC feature decoupling model, the HALN block clustering model, and the MVCL band selection model. First, in the FGMC feature decoupling model stage, $\mathcal{O}(B^2MN + B^3)$ and $\mathcal{O}(MN\log(MN))$ are needed to perform PCA and ERS processing on HSIs, respectively. Following this, the update of Z^i has the computational complexity of $\mathcal{O}(N^2M)$, and the update of C_X takes $\mathcal{O}(N^2)$. Then, in the HALN block clustering model stage, the time complexity of solving the nonlinear nearest-neighbor similarity matrix $\tilde{W}$ and the hybrid nearest-neighbor similarity matrix W is $2\mathcal{O}(N^2s)$. Finally, during the MVCL band selection model stage, it needs $\mathcal{O}(cN)$ and $\mathcal{O}(N^2c)$ to calculate $\tilde{P}_f$ and W_v, respectively. Specifically, the FGMC feature decoupling model takes $\mathcal{O}(B^3 + N^2M)$, the HALN block clustering model requires $2\mathcal{O}(N^2s)$, and the MVCL band selection model costs $\mathcal{O}(N^2c)$. When the number of iterations is T, the overall computational complexity of MMANP is $\mathcal{O}(B^3 + 2N^2s + N^2M + N^2c)T$. It should be noted that $B \ll MN$ in practice. The time complexity of different methods is presented in Table I.

To further ensure the applicability of MMANP in real-time or realistic scenarios, algorithm optimization can be considered from the following two perspectives.

1) Perform algorithmic optimization from the perspective of the optimization paradigm.
   a) In the FGMC feature decoupling model stage, representative samples are selected from all samples to serve as anchor samples for solving the Laplacian

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matrix. With this approach, the computational efficiency of the feature decoupling process can be improved.

b) In the HALN block clustering model stage, the $\delta\|\tilde{w}_i\|_2^2$ term in (16) can be removed to reduce the computational cost of solving the nonlinear nearest-neighbor similarity matrix. In summary, the aforementioned methods can reduce the computational cost of MMANP while keeping the computational accuracy.

2) Perform algorithmic optimization from the perspective of model architecture.

To further simplify the model structure, the HALN block clustering module can be replaced with less computationally expensive algorithms, such as CAN.

V. Experiments

In this section, extensive experiments are conducted to validate the effectiveness of the proposed MMANP on the four datasets. First, the four public datasets and the compared algorithms are introduced in the experiments. Then, the detailed setup of the experiments is presented. Next, the algorithms' experimental results are presented in detail and the superiority of MMANP is analyzed. Finally, the parameter sensitivity analysis, ablation study of MMANP, and visualization and analysis of results are given in detail.

A. Datasets

1) InP [51], [52]: This dataset was obtained by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1992. This image includes 145 × 145 pixels and 16 classes of features. With the removal of the interfered bands, here, we employ 200 bands for this study.

2) Salinas [52]: This dataset was obtained by the AVIRIS sensor over the Salinas image California, Salinas Valley. This image includes 512 × 217 pixels and 16 classes of features. With the removal of the interfered bands, here, we use 204 bands for this study.

3) Botswana [52]: This dataset was captured by the NASA EO-1 satellite sensor between 2001 and 2004. This image includes 1476 × 256 pixels and 14 classes of features. With the removal of the interfered bands, here, we adopt 145 bands for this study.

4) Pavia University (PU) [52]: This dataset was collected by the ROSIS sensor in 2002. This image includes 610 × 340 pixels and nine classes of features. With the removal of the interfered bands, here, we employ 103 bands for this study.

B. Compared Algorithms

To verify the effectiveness of the proposed MMANP, seven state-of-the-art band selection methods are used as compared algorithms for the experiments.

1) S4P [46]: An unsupervised self-representation band selection model preserves spectral–spatial structure information. The algorithm employs a multigraph fusion strategy for spatial structure preservation in the spectral space.

2) GRSC [44]: A graph-regularized spectral–spatial subspace clustering algorithm explores the obtained low-dimensional potential features and the similarity graph between these features in a new self-representation space for band selection.

3) HLFC [39]: A clustering-guided region-based hierarchical algorithm fuses the latent features extracted from each region through a hierarchical strategy for band selection work.

4) FNGBS [36]: A fast near-neighborhood grouping algorithm uses a coarse-fine strategy for band grouping and combines local density and information entropy to identify potential subsets of bands.

5) RDGSR [45]: A dual-graph self-representation algorithm employs a regularization term to constrain the self-representation model for band selection. Here, this model consists of the superpixel-based similarity graph and the similarity graph among bands.

6) RLFFC [53]: A regional feature fusion algorithm fuses region-based latent features to obtain shared latent feature representations for band selection.

7) EGCSR [35]: An efficient graph convolutional self-representation algorithm uses the structural information of HSIs for band selection.

C. Experimental Settings

To validate the performance of the proposed MMANP with other algorithms, k-nearest neighborhood (KNN), support vector machine (SVM), and random forest (RF) classifiers are employed to measure the selected bands of all algorithms. Random patches network (RPNet) [54], as one of the flexible deep learning methods, is now widely used in HSI classification. In this article, RPNet is used to evaluate the robustness of MMANP. The overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are used as evaluation metrics. Since the optimal number of selected bands for HSIs is still an open question, we choose the number of bands in the range of $\{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$ for different datasets and use different classifiers to classify them, respectively. The parameter k of the KNN classifier is set to 5. In the classification experiment, 10% of the samples in InP and Botswana images and 2% of the samples in Salinas and PU images are randomly selected as training samples, and the remaining samples are used as testing samples in the selected bands. To reduce the bias obtained by random sampling, each experiment is repeated ten times, and the average results are reported as the final classification results. Two parameters β and λ in MMANP are chosen from $\{0.001, 0.01, 0.1, 1, 10, 30, 50\}$, and the number of superpixel segmentations N_s is set to 10 for the InP, Botswana, and Salinas images, and 15 for the PU image. In the compared algorithms, we preserve the original authors' settings. The experiments are performed on a system with an NVIDIA GeForce GTX 3060 GPU and 48 GB of RAM.

D. Experimental Results

Table II shows the classification results of selected bands using different methods for the four datasets, where the top

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TABLE II
CLASSIFICATION RESULTS (%) OF ALL BAND SELECTION METHODS ON FOUR HYPERSPECTRAL DATASETS. THE OPTIMAL RESULTS ARE GIVEN IN BOLD

Dataset	Classifier	Metrics	S4P	GRSC	HLFC	FNGBS	RDGSR	RLFFC	EGCSR	MMANP
InP	KNN	OA	65.39	62.12	63.02	69.28	65.21	57.61	57.55	**85.99**
		AA	52.23	47.26	50.28	57.52	53.83	45.79	48.21	**74.85**
		Kappa	62.98	59.65	60.53	66.92	62.85	55.03	55.10	**84.52**
	SVM	OA	76.12	70.69	76.36	78.58	71.79	72.67	66.64	**91.87**
		AA	76.60	68.48	71.93	77.52	69.64	66.97	65.33	**87.09**
		Kappa	72.61	66.25	72.95	75.53	67.62	68.68	61.76	**90.73**
	RF	OA	69.29	64.86	71.39	71.36	65.63	68.43	59.90	**87.20**
		AA	56.60	50.45	58.82	58.12	52.91	56.01	52.00	**79.85**
		Kappa	66.91	62.31	69.09	69.05	63.16	66.01	57.36	**85.82**
	RPNet	OA	97.33	97.44	97.35	97.37	97.31	97.25	97.52	**97.70**
		AA	93.35	94.04	92.78	92.92	92.62	92.50	94.11	**94.21**
		Kappa	96.96	97.08	96.98	97.00	96.93	96.86	97.18	**97.38**
Salinas	KNN	OA	86.75	84.26	86.65	86.49	83.67	85.58	86.03	**92.42**
		AA	91.89	88.95	91.66	91.51	88.23	90.90	90.40	**92.33**
		Kappa	85.68	83.04	85.58	85.40	82.41	84.44	84.91	**91.72**
	SVM	OA	91.07	89.78	90.85	90.84	89.63	90.79	89.67	**96.13**
		AA	94.96	93.71	94.59	94.74	93.58	94.84	93.18	**96.44**
		Kappa	90.05	88.60	89.80	89.78	88.42	89.73	88.48	**95.69**
	RF	OA	87.36	85.39	86.98	87.24	84.22	86.76	86.38	**91.80**
		AA	92.06	89.58	91.45	91.76	88.66	91.43	90.81	**92.33**
		Kappa	86.34	84.25	85.94	86.21	83.02	85.70	85.29	**91.05**
	RPNet	OA	97.42	97.72	97.50	97.96	97.92	97.57	97.42	**98.32**
		AA	97.77	98.09	97.96	98.21	**98.29**	98.00	97.85	98.05
		Kappa	97.12	97.46	97.21	97.73	97.68	97.29	97.13	**98.13**
Botswana	KNN	OA	86.02	83.55	85.36	84.76	78.15	84.65	79.04	**95.14**
		AA	86.98	84.71	86.41	85.92	79.05	85.74	79.95	**94.41**
		Kappa	85.16	82.59	84.47	83.85	77.02	83.73	77.92	**94.77**
	SVM	OA	89.43	87.10	89.82	88.70	83.81	89.83	84.95	**98.61**
		AA	90.25	88.15	90.72	89.64	84.62	90.49	85.50	**98.42**
		Kappa	88.75	86.03	88.98	87.76	82.46	88.98	83.70	**98.29**
	RF	OA	84.26	81.91	83.76	83.00	77.29	83.10	78.68	**95.30**
		AA	85.10	82.77	84.56	83.91	77.69	83.91	79.43	**94.71**
		Kappa	83.33	80.90	82.81	82.02	76.12	82.12	77.55	**94.95**
	RPNet	OA	97.55	97.67	97.58	97.58	97.48	**97.83**	97.18	97.72
		AA	97.45	97.56	97.40	97.46	97.57	**97.62**	97.01	97.55
		Kappa	97.27	97.47	97.37	97.37	97.28	**97.65**	96.94	97.53
PU	KNN	OA	82.24	80.34	81.37	79.80	80.15	81.34	75.72	**91.87**
		AA	77.98	75.95	76.44	74.61	74.57	76.63	69.20	**87.02**
		Kappa	77.63	75.40	76.56	74.66	75.11	76.53	69.70	**89.42**
	SVM	OA	90.58	87.32	89.34	87.42	87.99	89.27	78.52	**95.09**
		AA	89.45	87.18	88.06	87.52	84.80	88.71	77.23	**93.33**
		Kappa	87.41	82.94	85.74	83.02	83.90	85.63	70.56	**93.48**
	RF	OA	83.55	81.57	82.45	82.07	82.40	82.28	77.44	**92.04**
		AA	78.50	76.56	76.95	76.83	76.07	76.94	70.89	**87.85**
		Kappa	79.23	76.88	77.88	77.45	77.84	77.67	71.95	**89.67**
	RPNet	OA	94.78	94.85	95.13	94.73	94.73	94.81	94.66	**96.57**
		AA	90.63	90.61	90.89	90.25	90.25	90.50	91.37	**92.15**
		Kappa	93.05	93.14	93.53	92.99	92.99	93.09	92.89	**95.43**

15 bands are selected as the band subset for each dataset. In Table II, the proposed MMANP achieves more satisfactory classification results. For example, the OA, AA, and Kappa results of MMANP are 20.60%, 22.62%, and 21.54% higher than the results of the S4P on the InP dataset with the KNN classifier, respectively. Similar experimental results are shown in the other three datasets. This fact not only shows the superior performance of MMANP but further demonstrates the importance of the FGMC feature decoupling model. In summary, the MMANP shows better robustness than other methods on these three datasets. To compare the MMANP with other methods as fully as possible, different numbers of bands are selected on different datasets for further validation. According to Fig. 6, we can summarize the following observations.

1) In most cases, the MMANP achieves more satisfactory classification results. For example, our method can outperform the compared methods by at least 16.71%, 13.29%, and 15.81% in the InP image by KNN, SVM, and RF classifiers. Notably, the selected bands of MMANP show satisfactory robustness in RPNet. These results demonstrate the superiority of our method. In other words, focusing on decoupling the bands is important because it emphasizes discriminative and important features and utilizes the hybrid similarity graphs to consider the distributions of spatial features in the Euclidean and non-Euclidean spaces simultaneously. To further analyze the selected bands, the selected 15 bands for different images using different methods are given in Fig. 7, respectively. In Fig. 7, the

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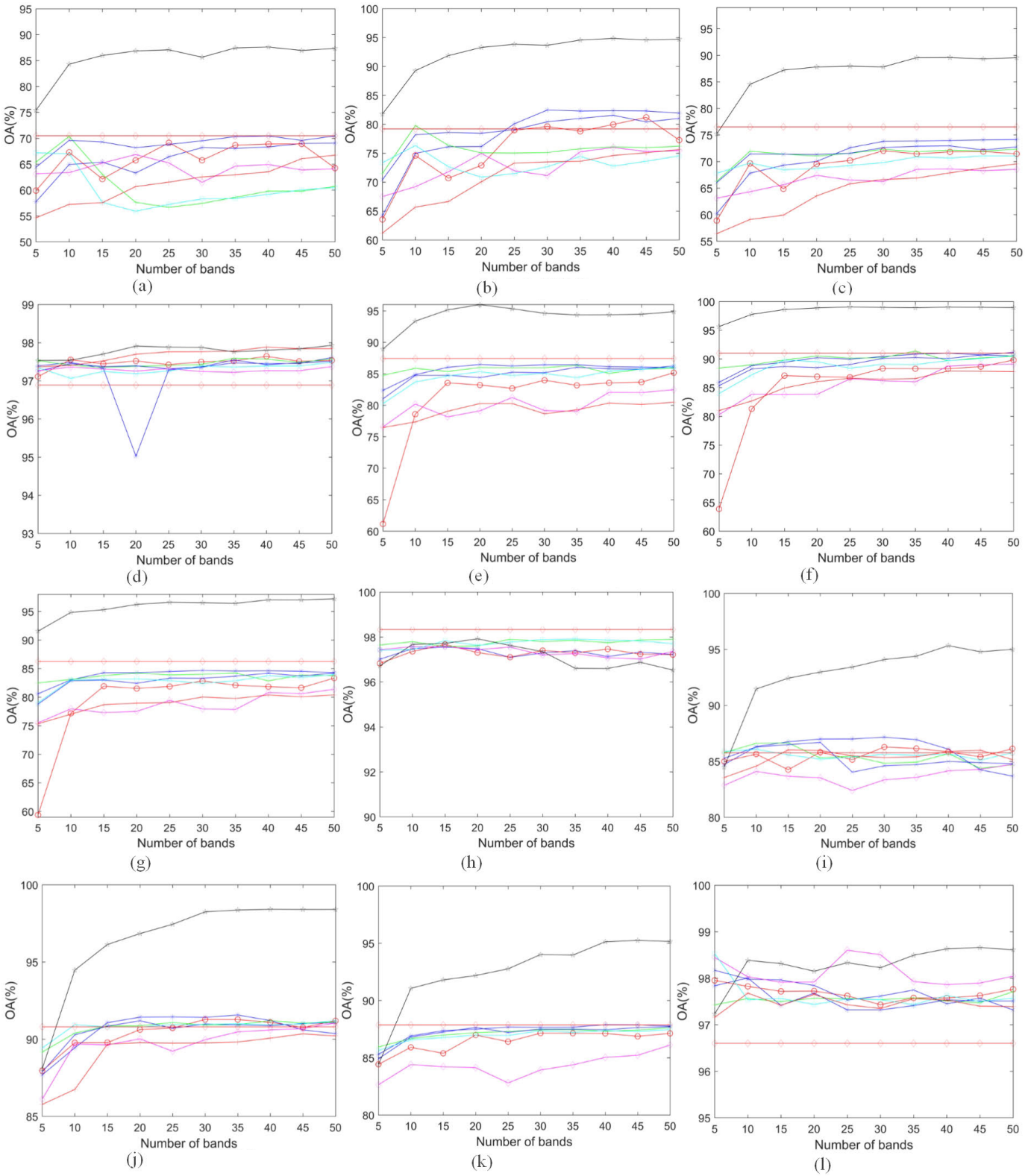


Fig. 6. OA results are obtained by various band selection methods with different numbers of selected bands on the InP, Botswana, Salinas, and PU datasets. (a) InP (KNN). (b) InP (SVM). (c) InP (RF). (d) InP (RPNet). (e) Botswana (KNN). (f) Botswana (SVM). (g) Botswana (RF). (h) Botswana (RPNet). (i) Salinas (KNN). (j) Salinas (SVM). (k) Salinas (RF). (l) Salinas (RPNet).

selected bands of the InP, Salinas, and PU images are more evenly distributed, while the selected bands of Botswana are distributed in the latter part of spectral bands. Here, some interesting latent regularities are found. The analysis of the features that contain fine-grained important information within the

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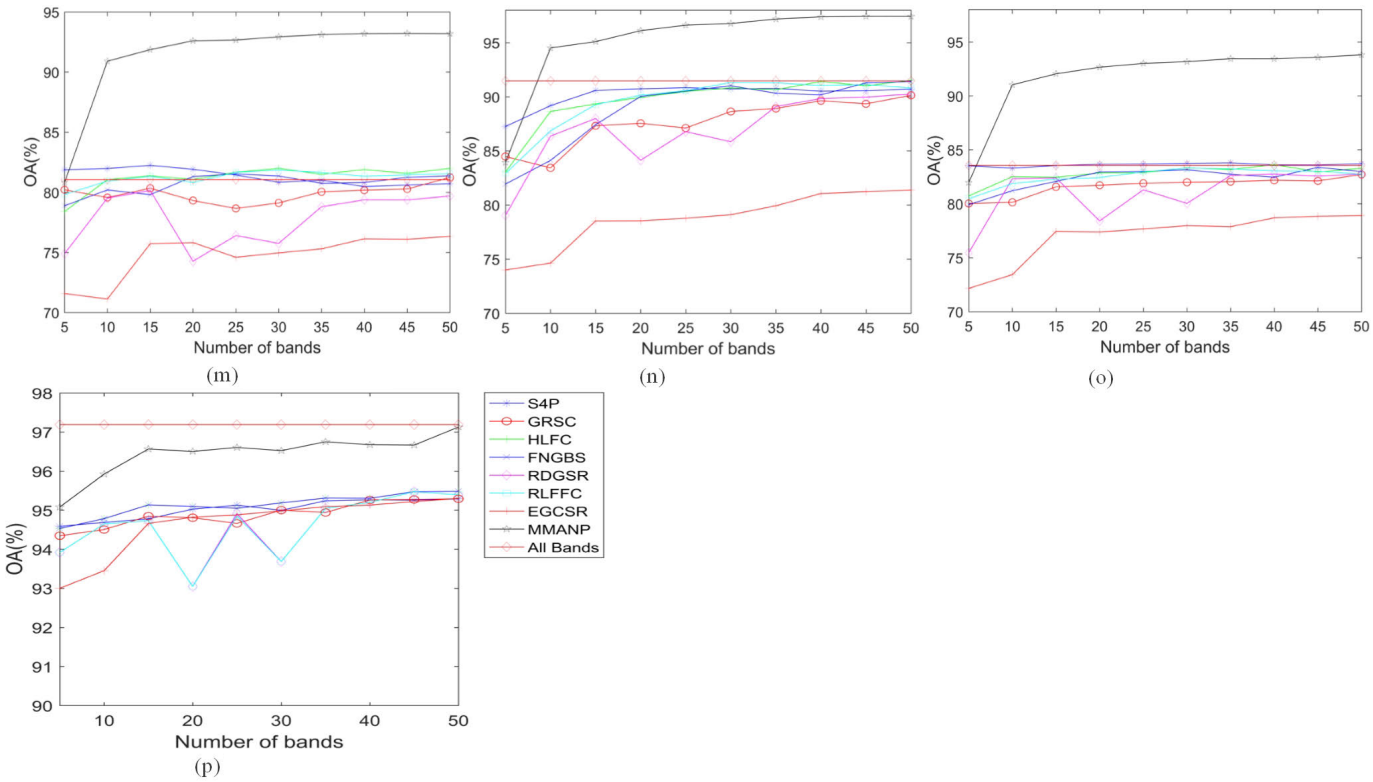


Fig. 6. *(Continued.)* OA results are obtained by various band selection methods with different numbers of selected bands on the InP, Botswana, Salinas, and PU datasets. (m) PU (KNN). (n) PU (SVM). (o) PU (RF). (p) PU (RPNet).

bands facilitates the selection of more discriminative bands.

2) Among all competitors, our method significantly improves the classification performance even with a small number of selected bands. The reasons may include the following two aspects of this phenomenon.
   a) Design of the FGMC feature decoupling model in MMANP.
      To more accurately delineate the intrinsic information of each band in HSIs, the FGMC model can express discriminative and important features with the fine-grained manifold structures of bands. With this approach, it is beneficial to represent the intrinsic discriminative features in bands. Furthermore, this factor contributes to guaranteeing the discriminative quality of the selected bands.
   b) The developed MVCL feature selection model in MMANP.
      To alleviate the limitations of the selected bands in a single-view perspective, the MVCL model performs adaptive band selection for each subset separately from the multiview cross-granularity perspective. It is worth mentioning that the MVCL model first adaptively determines the number of bands for different subsets according to the contributions of band subsets. Then, the bands with high contribution are selected as the selected bands in each subset adaptively. By employing this approach, it is beneficial to improve the reliability of the selected bands.

   Furthermore, there is not much change in classification accuracy after ten bands in Fig. 6. The reason for this phenomenon may be the adaptive band selection performed by the MVCL model of MMANP. That is, the selected bands by MMANP are determined according to the contributions of each band and each subset together, and the bands with large contributions in the MVCL model are selected with priority. In other words, as the number of selected bands increases, bands with smaller contributions are gradually selected.
   Using the Salinas image as an example, compared to the S4P, the OA improves by 5.67%, 5.06%, 4.44%, and 0.90% when selecting 15 bands in Fig. 6, respectively. Similarly, AA and Kappa also improved, respectively. Thus, these results further demonstrate the superiority of MMANP.
3) The MMANP can outperform S4P, which employs the Gaussian kernel function to supplement the computation of similarity in the segmentation regions. Since the Gaussian kernel function is difficult to effectively represent the complex distributional properties of spatial information, the resulting similarity matrix has limited performance.
   To visualize the validity of MMANP, the classification maps of the selected five bands are shown in different images in Figs. 8–11. In Figs. 8–11, MMANP can more smoothly classify the ground classes of dif-

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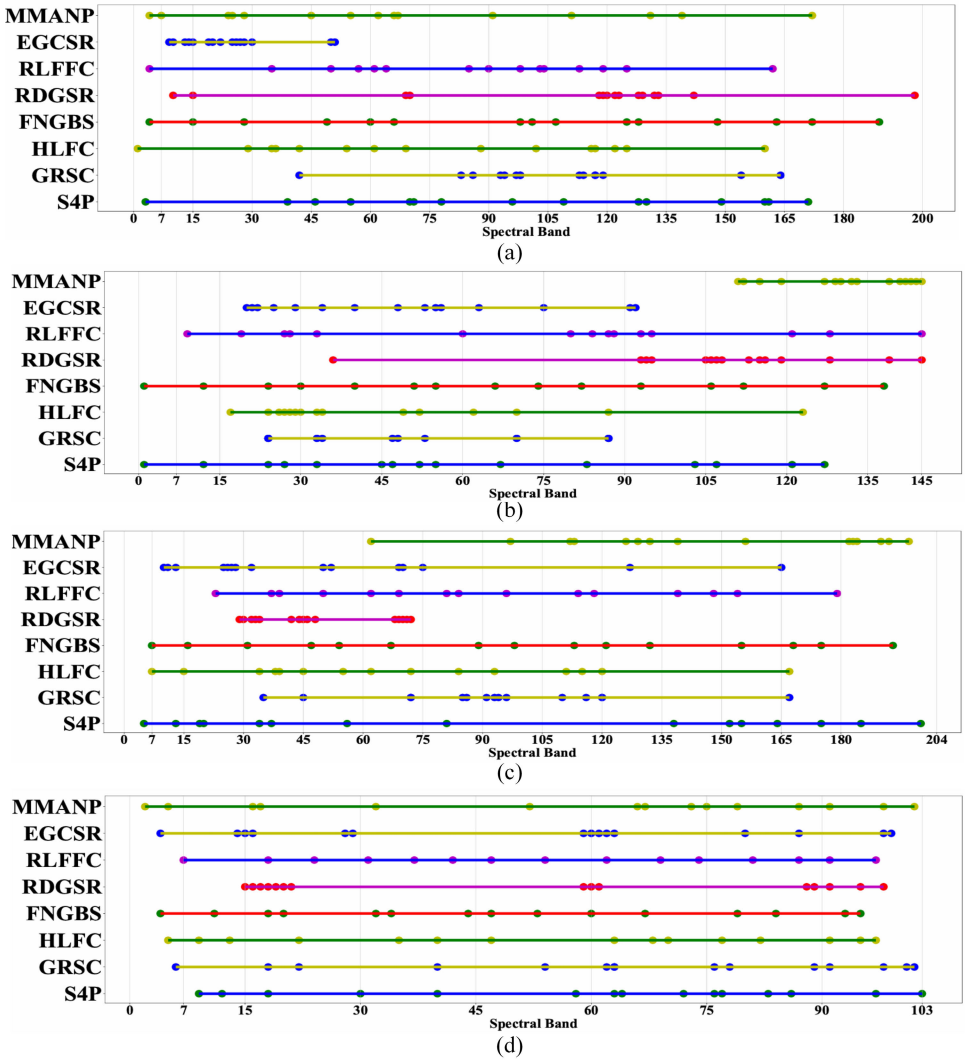


Fig. 7. Top 15 spectral bands are selected by different methods. (a) InP. (b) Botswana. (c) Salinas. (d) PU.

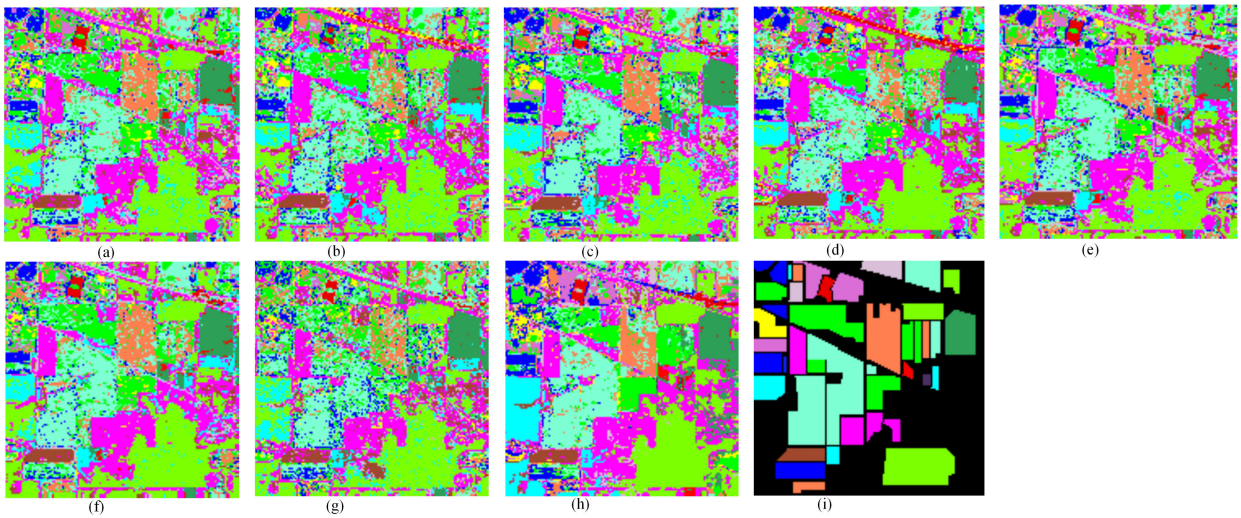


Fig. 8. Classification maps for the top five bands are selected by different band selection methods in the InP image. (a) S4P. (b) GRSC. (c) HLFC. (d) FNGBS. (e) RDGSR. (f) RLFFC. (g) EGCSR. (h) MMANP. (i) Ground truth.

ferent scenes. This fact demonstrates the importance of MMANP to decouple the bands into multigranularity features. This intriguing finding provides a direction for further research on how to utilize the advantages of band decoupling in band selection for HSIs.

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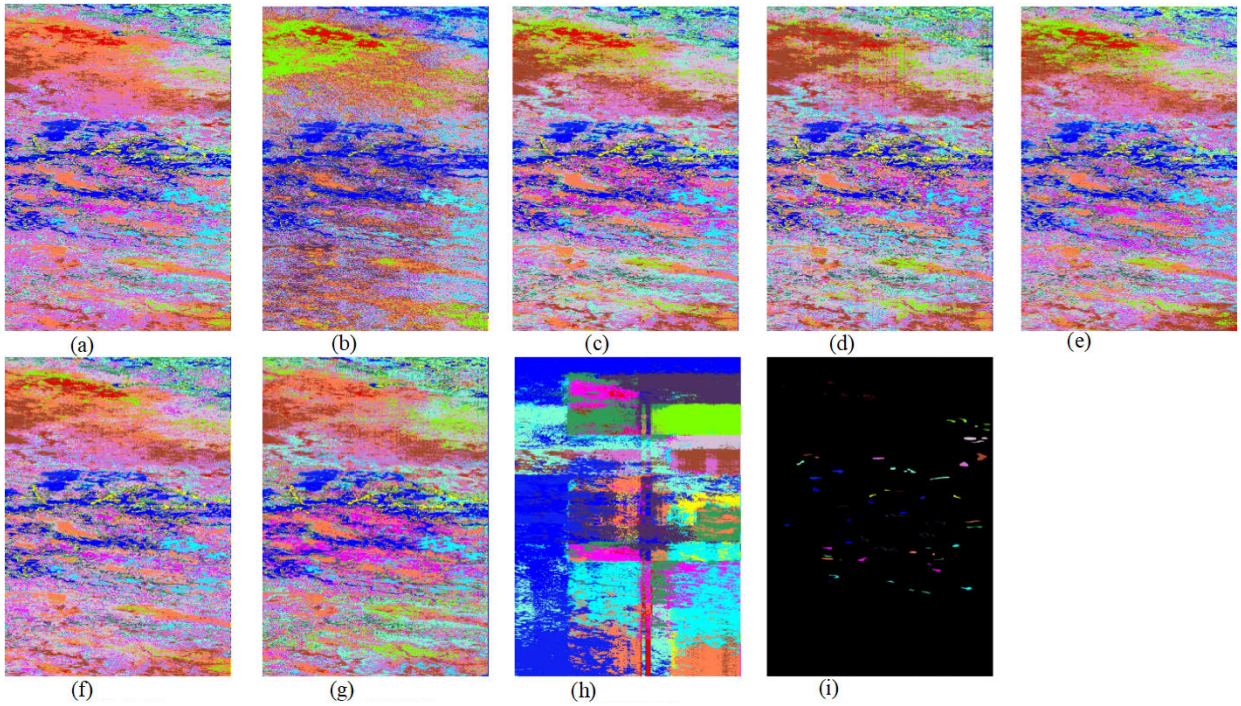


Fig. 9. Classification maps for the top five bands are selected by different band selection methods in the Botswana image. (a) S4P. (b) GRSC. (c) HLFC. (d) FNGBS. (e) RDGSR. (f) RLFFC. (g) EGCSR. (h) MMANP. (i) Ground truth.

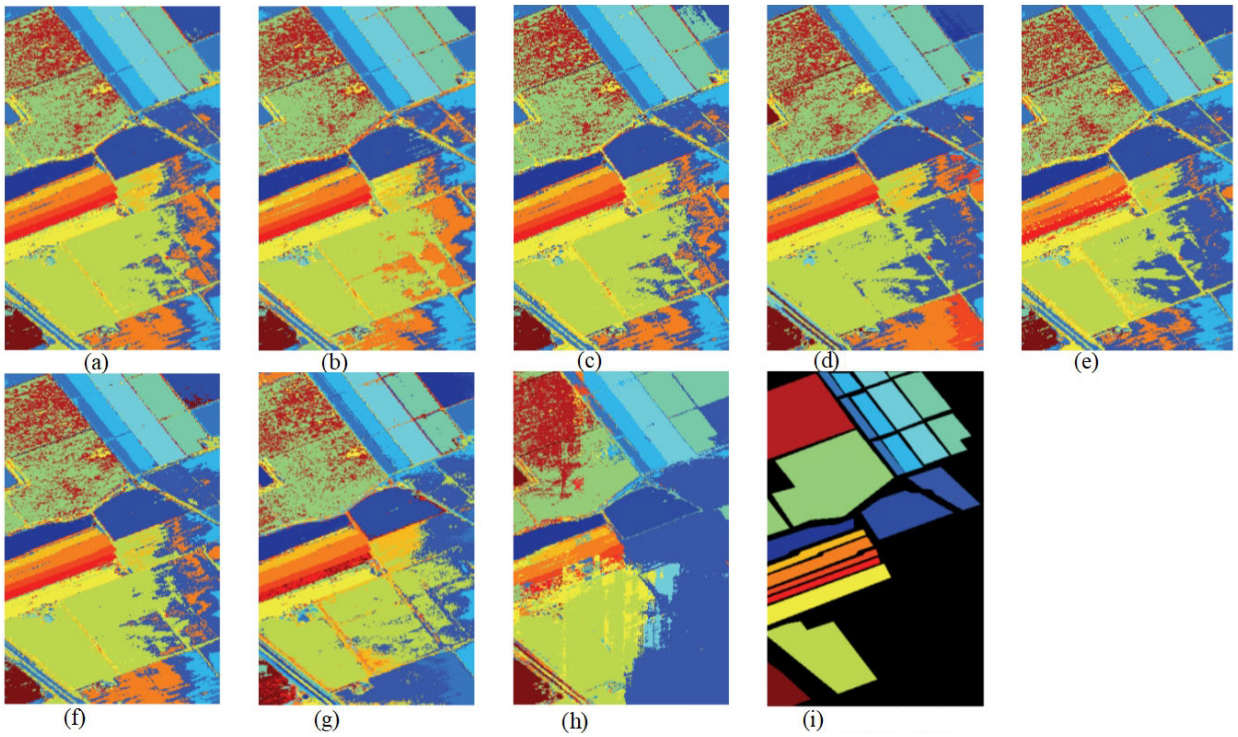


Fig. 10. Classification maps for the top five bands are selected by different band selection methods in the Salinas image. (a) S4P. (b) GRSC. (c) HLFC. (d) FNGBS. (e) RDGSR. (f) RLFFC. (g) EGCSR. (h) MMANP. (i) Ground truth.

E. Discussion

1) Ablation Study: To validate the effectiveness of the components in MMANP, we perform ablation experiments to further understand the effectiveness of the FGMC model, the HALN block clustering model, and the MVCL model.

a) Effect of FGMC model: The band information of the HSI is represented as accurately as possible. In Fig. 12(a), the experimental results are presented to verify the validity of the FGMC model. It can be intuitively shown that MMANP with the FGMC feature decoupling model has better

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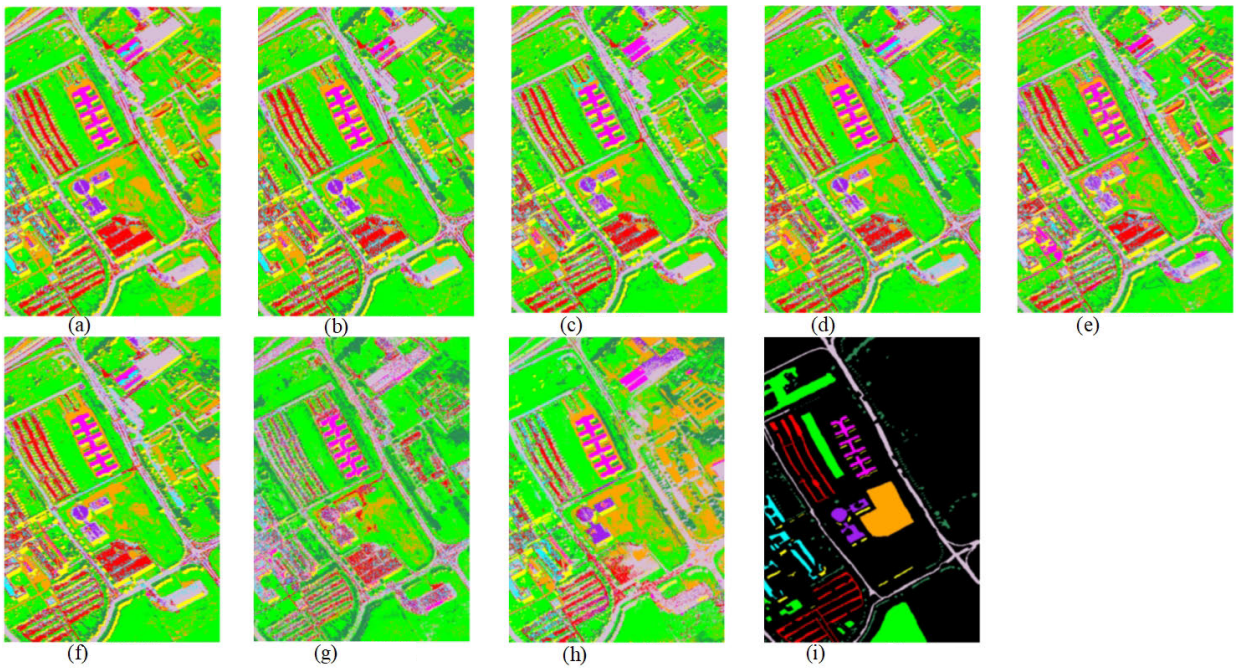


Fig. 11. Classification maps for the top five bands are selected by different band selection methods in the PU image. (a) S4P. (b) GRSC. (c) HLFC. (d) FNGBS. (e) RDGSR. (f) RLFFC. (g) EGCSR. (h) MMANP. (i) Ground truth.

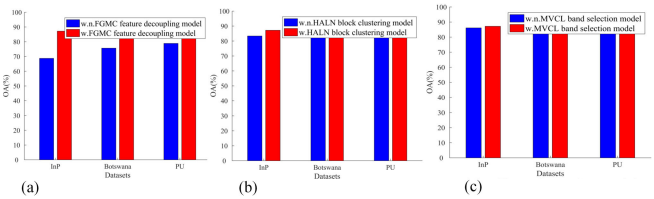


Fig. 12. Ablation study of MMANP with different components on HSIs. (a) Effect of FGMC model. (b) Effect of HALN model. (c) Effect of MVCL model.

experimental results. This is attributed to the fact that the FGMC model decouples the band information of HSIs into important and discriminative features for further analysis.

b) Effect of HALN model: The similarity graph among bands is used to cluster the bands of HSIs. In Fig. 12(b), the results of related experiments are presented to explore the effectiveness of HALN. It can be intuitively seen that MMANP with the hybrid nearest-neighbor similarity graph outperforms MMANP considering only the nearest-neighbor similarity graph in the Euclidean space. This factor demonstrates the effectiveness of HALN.

c) Effect of MVCL model: The design of the band selection model from the multiview cross-granularity perspective. In Fig. 12(c), we present the experimental results to verify the validity of the MVCL model. It can be observed intuitively that the MMANP with the MVCL model outperforms the MMANP with only the multiview model.

2) Parameter Sensitivity Analysis: In this article, there are two important parameters that need to be set, i.e., parameters β and λ. To verify the parameter sensitivity effect of the regularization parameter on the selected bands, we, respectively, fix β and λ to select different numbers of bands. The experimental results are presented in Fig. 13. In Fig. 13, the effect of different λ is small for different numbers of selected bands of HSIs. Similarly, the results fluctuate less for different β for different numbers of selected bands.

Fig. 14 shows the degree of changes of MMANP with different parameters, which further visualizes the degree of variation in MMANP for different parameters in Fig. 13. Here, "di" is the difference in band selection performance between group "i" and group "$i+1$." In Fig. 14, the performance of MMANP fluctuates less in different images. This phenomenon indicates that the performance of MMANP is stable with different regularization parameters. That is, MMANP can obtain satisfactory stabilization results in a wide range. To further ensure the validity and reliability of the FGMC feature decoupling model as much as possible, here, we consider the CF of multigranularity important and discriminative features and the preservation of the manifold structure simultaneously. Thus, here, let $\lambda, \beta = 1$. In summary, it can be seen that MMANP has satisfactory stability in Figs. 13 and 14.

3) Visualization and Analysis of Results: To visualize the learning ability in the subspace of the MMANP, we, respectively, project the selected 15 bands by different models into intuitive 2-D subspaces obtained by PCA. To clearly show the comparative results of the selected bands, we compare the two datasets as an example. Specifically, the InP and Botswana datasets include 16 and 14 classes of samples, respectively. For ease of record keeping, we abbreviate "class i" as "i." The visual comparison results are shown in Figs. 15 and 16.

Figs. 15 and 16 show the distributions of low-dimensional subspace features learned by different models. The MMANP can discriminate between similar samples as much as possible, while other methods cannot separate the various classes of samples in the InP dataset. Similarly, the same conclusion

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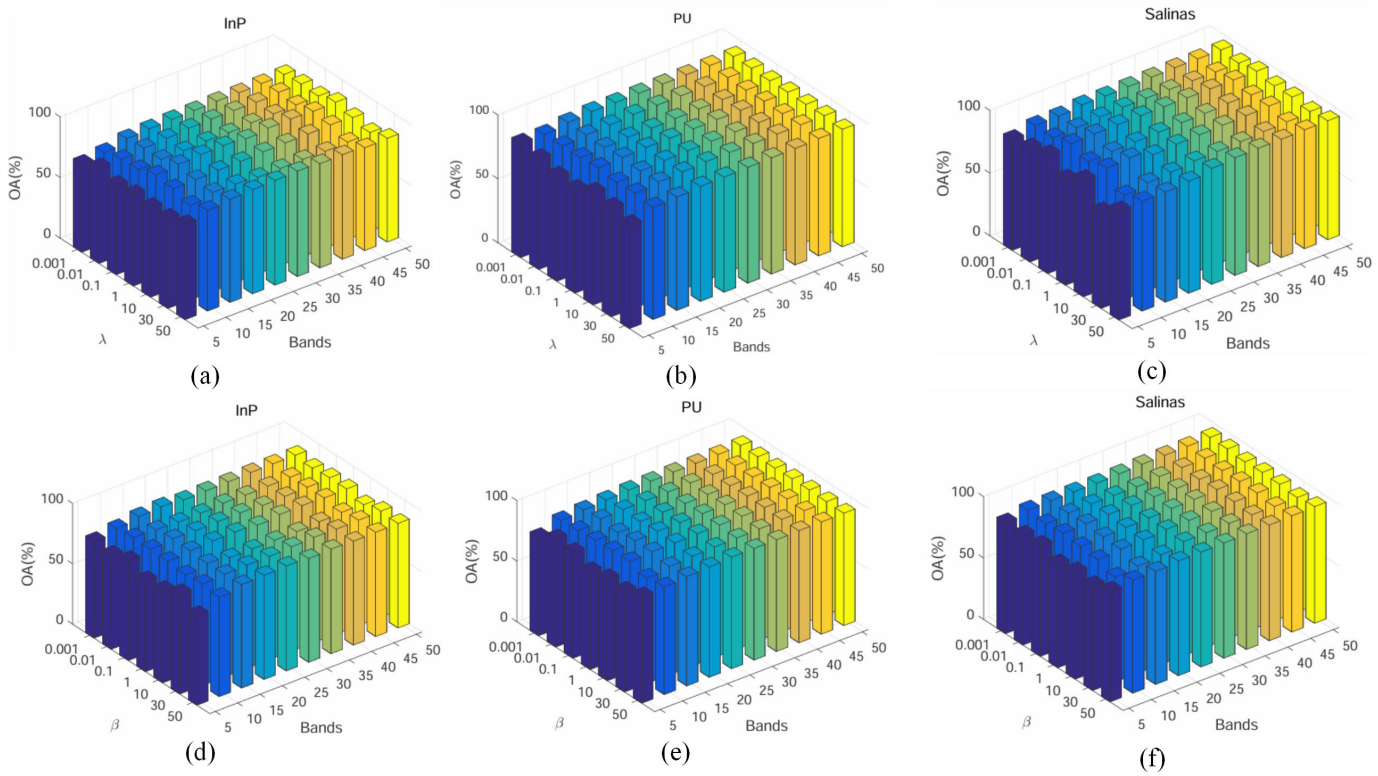


Fig. 13. Performance variation of MMANP for different parameters. (a) and (d) InP. (b) and (e) PU. (c) and (f) Salinas.

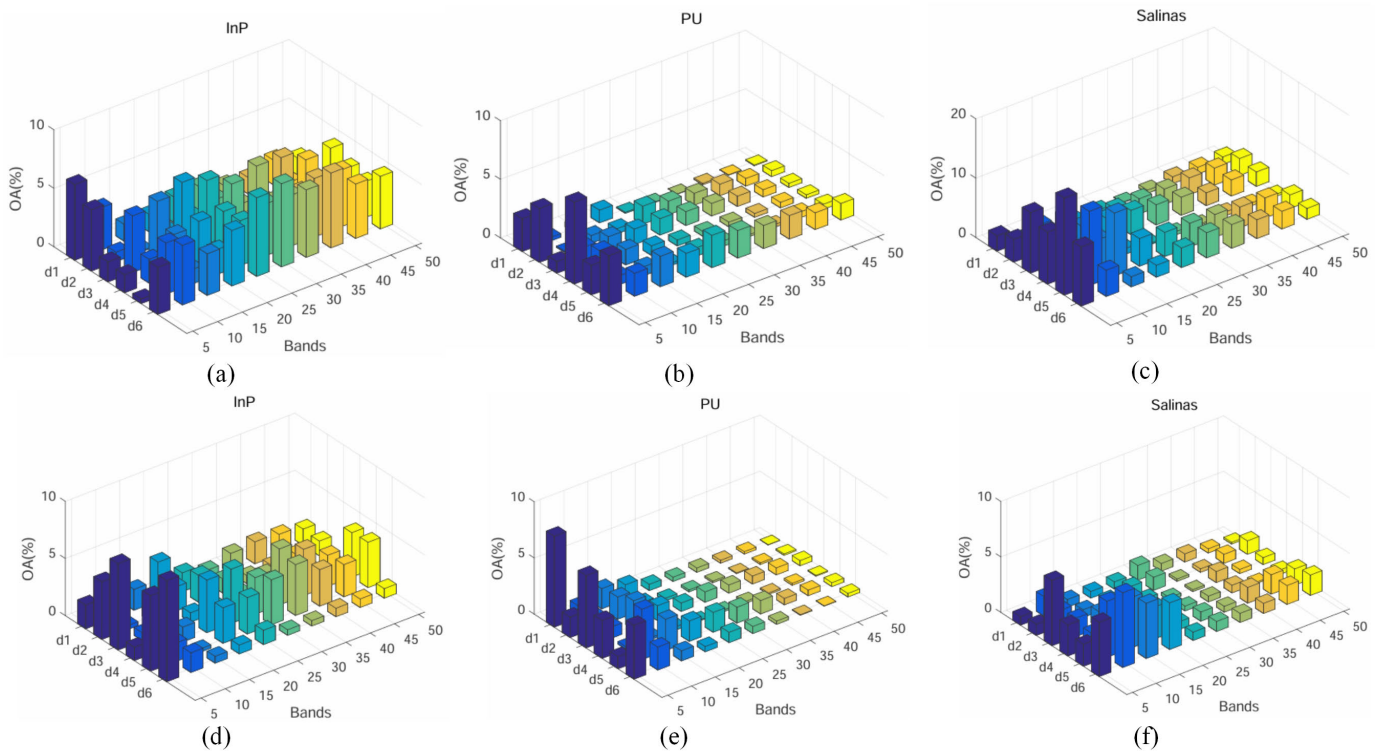


Fig. 14. Degree of changes of MMANP with different parameters. (a) and (d) InP. (b) and (e) PU. (c) and (f) Salinas.

can be derived from the Botswana image. This fact intuitively demonstrates that the selected bands by the MMANP improve the expressiveness of discriminative features by introducing pseudo-class label constraints.

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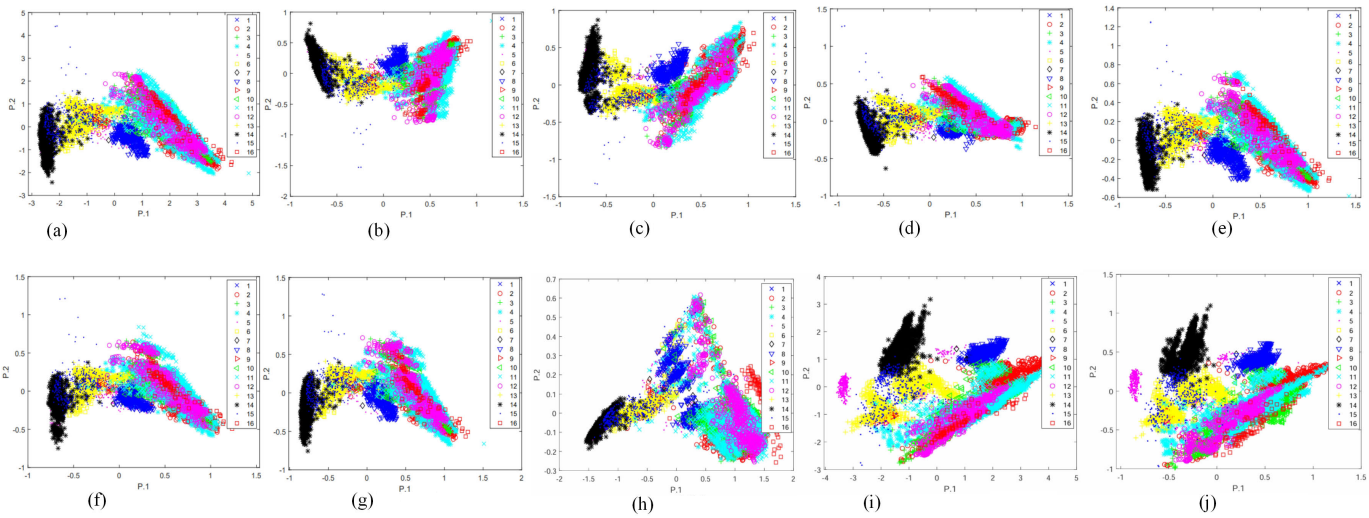


Fig. 15. Visualization of results with different band selection methods for the InP dataset. (a) Original InP image. (b) S4P. (c) GRSC. (d) HLFC. (e) FNGBS. (f) RDGSR. (g) RLFFC. (h) EGCSR. (i) Enhanced InP image. (j) MMANP.

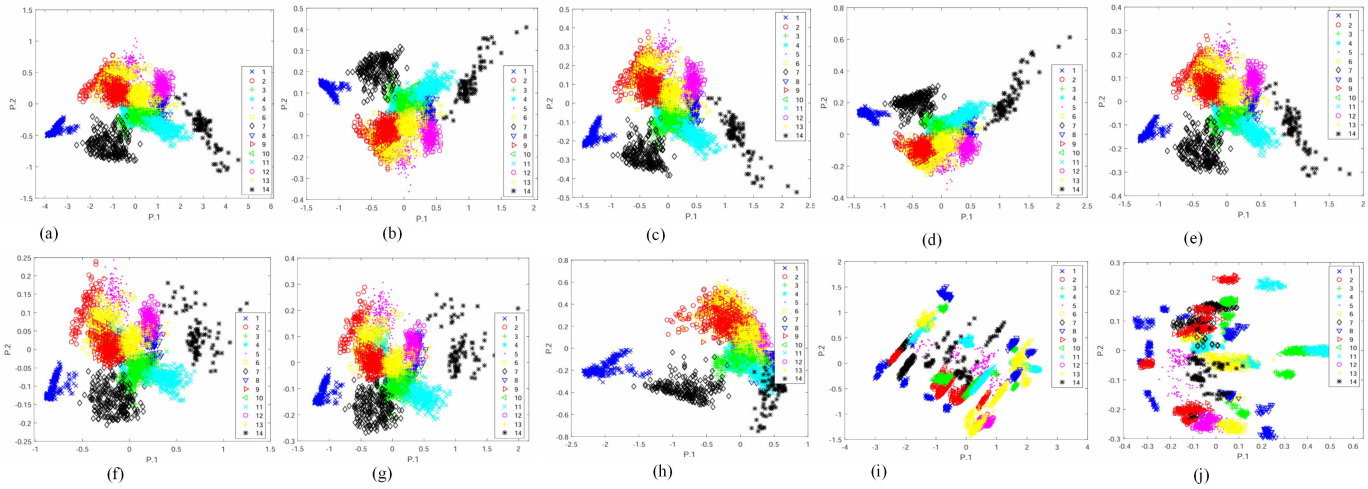


Fig. 16. Visualization of results with different band selection methods for the Botswana dataset. (a) Original Botswana image. (b) S4P. (c) GRSC. (d) HLFC. (e) FNGBS. (f) RDGSR. (g) RLFFC. (h) EGCSR. (i) Enhanced Botswana image. (j) MMANP.

VI. Conclusion

In this article, a novel MMANP method is proposed for band selection. Specifically, we first transform the problem of analyzing the intrinsic discriminative components of bands into a feature decoupling problem and propose an FGMC feature decoupling model. This model can obtain spatially discriminative and important features with fine-grained manifold structures in bands. Additionally, to overcome the limitation of considering only the distribution of spatial features of each band in the Euclidean or non-Euclidean space, the HALN block clustering model is proposed. This model takes the nonlinear and linear nearest-neighbor similarity graphs for each band to be considered together in the unified model to assist the more accurate clustering among similar bands. Next, the MVCL feature selection model is proposed to overcome the limitation of the single-view perspective. This model harmoniously integrates information at different levels of granularity in HSIs into a unified framework to perform adaptive band selection for each cluster. Extensive experimental results on four datasets demonstrate that the proposed MMANP outperforms some state-of-the-art methods.

Although the MMANP method can perform well in the band selection tasks of HSIs, it can be further improved in the following directions in the future. In this article, adaptively determining the optimal number of selected bands is still an open problem. In further work, we will perform the band selection for HSIs from the following considerations. First, we attempt to consider the spectral–spatial information of HSIs jointly to extract a more discriminative subset of bands. Then, we attempt to combine search algorithms to adaptively determine the optimal number of selected bands.

Appendix
Solution to (30)

Equation (30) is defined as

$$\min_{\tilde{w}_{ij}} \sum_{j=1}^{B} \left\|\tilde{z}_i - \tilde{z}_j\right\|_2^2 \tilde{w}_{ij} + \delta \left\|\tilde{w}_i\right\|_2^2$$

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$$+\sum_{j=1}^{B}\left(\left\|\tilde{w}_i\hat{z}_i-\tilde{w}_i\hat{z}_j\right\|_2^2 w_{ij}\right)$$
$$\text{s.t. } 1^{\mathrm{T}}\tilde{w}_i=1,\quad 0\le\tilde{w}_{ij},\ \tilde{w}_{ii}=0. \tag{42}$$

Here, let $d_{ij}=\|\tilde{z}_i-\tilde{z}_j\|_2^2$. Thus, (42) can be written as

$$\min\frac{1}{2}\left\|\tilde{w}_i+\frac{d_i}{2\delta}\right\|_2^2+\frac{1}{2\delta}\left\|\tilde{w}_i\hat{z}_i-\tilde{w}_i\hat{z}_j\right\|_2^2 w_{ij}$$
$$\text{s.t. } \tilde{w}_{ii}=0,\quad \tilde{w}_{ij}\ge 0,\ 1^{\mathrm{T}}\tilde{w}_i=1. \tag{43}$$

The Lagrangian function of (43) can be calculated

$$\mathcal{L}(\tilde{w}_i,\eta,\xi)=\frac{1}{2}\left\|\tilde{w}_i+\frac{d_i}{2\delta}\right\|_2^2+\frac{1}{2\delta}\left\|\tilde{w}_i\hat{z}_i-\tilde{w}_i\hat{z}_j\right\|_2^2 w_{ij}-\eta\left(1^{\mathrm{T}}\tilde{w}_i-1\right)-\xi\tilde{w}_i \tag{44}$$

where η and ξ are the Lagrangian coefficient scalar and the Lagrangian coefficient vector, respectively. To solve (44), the derivative with respect to $\tilde{w}_{ij}$ is calculated by

$$\tilde{w}_{ij}+\frac{d_{ij}}{2\delta}+\frac{1}{\delta}w_{ij}\left(\tilde{w}_{ij}\hat{z}_i-\tilde{w}_{ij}\hat{z}_j\right)^{\mathrm{T}}\left(\hat{z}_i-\hat{z}_j\right)-\eta-\xi_j=0. \tag{45}$$

According to KKT conditions, $\tilde{w}_{ij}$ can be formulated by

$$\tilde{w}_{ij}=\left(\frac{-\frac{d_i}{2\delta}+\eta}{1+\frac{1}{\delta}w_{ij}\left(\hat{z}_i-\hat{z}_j\right)^{\mathrm{T}}\left(\hat{z}_i-\hat{z}_j\right)}\right)_+. \tag{46}$$

Here, we constrain $\|\tilde{w}_i\|_0=k$, i.e., $\tilde{w}_{ik}>0$ and $\tilde{w}_{i,k+1}=0$. Thus, we can obtain the mathematical expression as follows:

$$-\frac{d_{ik}}{2\delta}+\eta>0\quad\text{and}\quad-\frac{d_{i,k+1}}{2\delta}+\eta\le 0. \tag{47}$$

Next, according to (46) and the constraint $1^{\mathrm{T}}\tilde{w}_i=1$, the mathematical expression for η is

$$\eta=\frac{1}{k}\left[1+\frac{\sum_{h=1}^{k}d_{ih}}{2\delta}+\sum_{h=1}^{k}\frac{w_{ij}}{\delta}\left(\hat{z}_i-\hat{z}_j\right)^{\mathrm{T}}\left(\hat{z}_i-\hat{z}_j\right)\right]. \tag{48}$$

According to (47) and (48), we have

$$\frac{-\sum_{h=1}^{k}d_{ih}+kd_{ik}-2\sum_{h=1}^{k}w_{ih}\left(\hat{z}_i-\hat{z}_j\right)^{\mathrm{T}}\left(\hat{z}_i-\hat{z}_j\right)}{2}<\delta$$
$$\le\frac{-\sum_{h=1}^{k}d_{ih}+kd_{i,k+1}-2\sum_{h=1}^{k}w_{ih}\left(\hat{z}_i-\hat{z}_j\right)^{\mathrm{T}}\left(\hat{z}_i-\hat{z}_j\right)}{2}. \tag{49}$$

Finally, let $\hat{Z}_{ij}=(\hat{z}_i-\hat{z}_j)^{\mathrm{T}}(\hat{z}_i-\hat{z}_j)$, and the optimal solution of $\tilde{w}_{ij}$ can be denoted as

$$\tilde{w}_{ij}=\begin{cases}\dfrac{d_{i,k+1}-d_{ij}}{-\sum_{h=1}^{k}d_{ih}+kd_{i,k+1}-2\sum_{h=1}^{k}w_{ih}\hat{Z}_{ij}+2w_{ij}\hat{Z}_{ij}}, & j\le k\\ 0, & j>k.\end{cases} \tag{50}$$

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